

CAREX – Comprehensive Adaptive Rehabilitation Expert AI-Powered Personalized Rehabilitation Platform

PHASE I REPORT

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ANNA UNIVERSITY, CHENNAI
BONAFIDE CERTIFICATE

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ABSTRACT

CAREX (Comprehensive Adaptive Rehabilitation Expert) is an AI-powered personalized rehabilitation platform developed to support patients during their recovery process. Traditional rehabilitation often depends on periodic clinical visits and provides limited monitoring of patients while exercising at home. CAREX addresses this gap by providing personalized rehabilitation plans based on the patient's condition, recovery stage, and treatment history.

The system uses computer vision and MediaPipe Pose to monitor exercises, analyze body movements, evaluate exercise form, and count repetitions. It also uses AI-based recovery analysis to track patient progress and provide insights into recovery outcomes. Healthcare professionals can remotely review patient progress through a dedicated dashboard and modify rehabilitation plans when required. Additional features such as medication and appointment reminders support better adherence to the rehabilitation process. An optional ESP32 and MPU6050-based wearable module can assist with movement and fall detection.

CAREX aims to provide a continuous, personalized, and technology-assisted rehabilitation experience while supporting healthcare professionals in monitoring patients remotely.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CAREX	Comprehensive Adaptive Rehabilitation Expert
DFD	Data Flow Diagram
ESP32	Espressif Systems 32-bit Microcontroller
IMU	Inertial Measurement Unit
ML	Machine Learning
MPU6050	Motion Processing Unit 6050
ROM	Range of Motion
SpO ₂	Peripheral Oxygen Saturation
UML	Unified Modeling Language
UI	User Interface
XGBoost	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Rehabilitation is an important part of recovery for patients who have undergone surgery, experienced fractures, sports injuries, stroke or other physical conditions. Recovery does not depend only on medical treatment; patients also need to perform prescribed exercises regularly, follow treatment instructions and attend scheduled consultations. However, maintaining this process outside the clinical environment can be difficult because healthcare professionals cannot continuously monitor every patient's activities.

In many cases, patients perform rehabilitation exercises at home without immediate feedback regarding their posture, movement or consistency. Progress may only be reviewed during periodic consultations, making it difficult to identify problems at an early stage. Patients may also forget medications, appointments or scheduled exercises, which further affects the overall outcome of the rehabilitation process.

With the development of Artificial Intelligence, computer vision, machine learning and cloud-based healthcare technologies, it is possible to provide more continuous support during rehabilitation. CAREX (Comprehensive Adaptive Rehabilitation Expert) is proposed as an AI-powered personalized rehabilitation platform that connects patients and healthcare professionals through a unified digital environment, using patient-specific information such as medical condition, age, recovery stage and treatment history to support personalized rehabilitation and recovery tracking.

1.2 OBJECTIVE

The main objective of this project is to develop an AI-powered rehabilitation platform that provides personalized, continuous and data-driven rehabilitation support while allowing healthcare professionals to remotely monitor patient progress. The specific objectives of the system are:

- To create a personalized rehabilitation profile for every patient.
- To provide rehabilitation exercises according to the patient's condition and recovery stage.

- To evaluate exercise performance using computer vision.
- To count exercise repetitions automatically and identify incorrect exercise posture or form.
- To track rehabilitation progress over time, including range of motion, adherence and pain trend.
- To monitor relevant patient health information such as heart rate and SpO₂.
- To use machine learning for recovery and readiness analysis.
- To provide medication, hydration and appointment support.
- To provide an AI-powered rehabilitation assistant for patient queries.
- To allow doctors and healthcare professionals to monitor patients remotely and modify treatment plans.
- To provide reports and useful recovery analytics to support clinical decision-making.

1.3 EXISTING SYSTEM

The existing rehabilitation process can generally be divided into the following approaches.

1.3.1 In-clinic Physiotherapy

The patient visits a hospital or clinic and the physiotherapist observes the exercises directly. Supervision is available only during the scheduled appointment, and the physiotherapist has no visibility into how the patient performs exercises between visits.

1.3.2 Paper-based or Verbal Instructions

The patient receives exercise instructions on paper or verbally and performs them at home. There is no automatic feedback about whether the exercise is being performed correctly, and adherence depends entirely on the patient's own discipline and memory.

1.3.3 General Fitness Applications

Fitness applications can track workouts, repetitions and general activity levels. However, they are not designed around a patient's specific injury, medical history or rehabilitation stage, and therefore cannot substitute for a rehabilitation-specific system.

1.3.4 Telehealth / Video Consultation

A patient can communicate with a physiotherapist through a video call, which provides remote interaction. However, this is still largely periodic, depends on both parties being

available at the same time, and does not continuously collect rehabilitation information between consultations.

Limitations of Existing Systems

- Periodic, appointment-based monitoring rather than continuous supervision.
- No automatic, real-time feedback on exercise form or technique.
- Limited personalization to the patient's specific condition and recovery stage.
- No integrated recovery prediction or readiness analysis.
- Poor adherence monitoring for exercises, medication and appointments.
- Separate, disconnected tools for different rehabilitation-related activities.

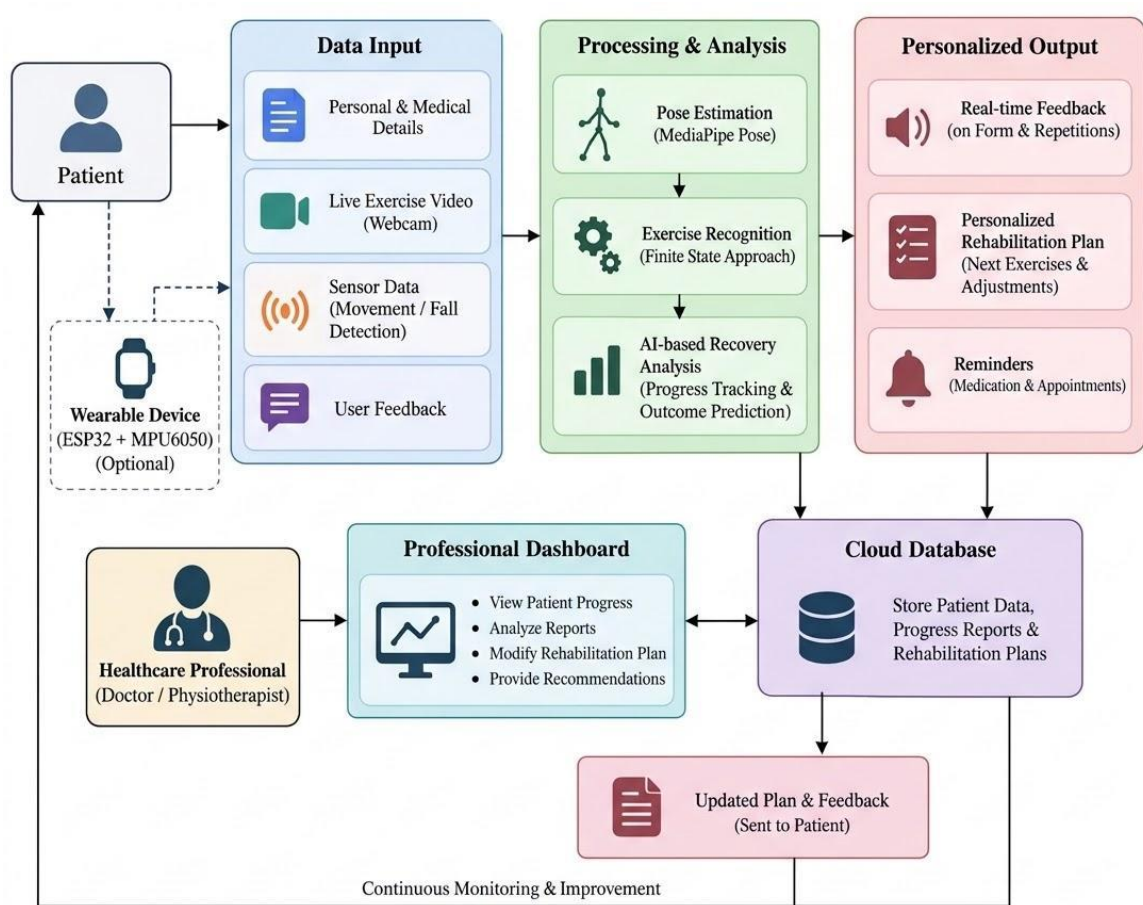
1.4 PROPOSED SYSTEM

CAREX is proposed as an AI-powered personalized rehabilitation platform that integrates patient assessment, rehabilitation planning, exercise monitoring, progress tracking, recovery analysis and healthcare-professional supervision within a single system. Rather than treating rehabilitation as a set of disconnected tools, CAREX brings together a patient application, an AI/processing engine and a doctor dashboard into one coordinated platform.

The system creates a personalized rehabilitation profile for each patient based on their medical condition, age, recovery stage and treatment history. During exercise sessions, a computer-vision component analyzes the patient's body movement through a smartphone camera, calculates joint angles, counts repetitions and evaluates exercise form. This is complemented by an optional wearable extension (ESP32 with an MPU6050 IMU) for situations where camera-based measurement may be affected by lighting or occlusion.

Rehabilitation progress is tracked using indicators such as range of motion, adherence rate, form compliance and pain trend. A machine-learning component, built around XGBoost, supports recovery analysis by estimating recovery duration, identifying slow-recovery risk and suggesting the appropriate next rehabilitation stage. An AI rehabilitation assistant helps patients with day-to-day queries, while medication, hydration and appointment reminders support overall adherence. Throughout, the AI components are designed to provide analytical support and recommendations, while healthcare professionals remain responsible for reviewing and modifying the treatment plan.

The overall proposed flow of the system can be summarized as:



Overall Proposed System Flow of CAREX

Fig. 1.1 Overall Proposed System Flow

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

2.1.1 Telerehabilitation for Stroke Patients

Laver et al. studied telerehabilitation for stroke patients and found that remote rehabilitation can be useful and can produce outcomes comparable to conventional, in-clinic rehabilitation in some situations. However, the approach does not by itself provide the complete personalization and continuous intelligent analysis that CAREX aims to provide.

2.1.2 AI-based Rehabilitation Technologies

Sumner et al. reviewed AI-based rehabilitation technologies, including mobile applications, robotics, gaming and wearable devices. The review also identified challenges such as reliability, the technology skills required of patients, user fatigue, and limited clinical evaluation of these systems.

2.1.3 Personalized Exercise Recommendation

Kim et al. investigated personalized exercise recommendation using a graph neural network, showing that different patient characteristics can be used to recommend more suitable exercises. However, the approach focuses mainly on recommendation rather than complete rehabilitation management.

2.1.4 Machine-Learning-based Recovery Prediction

Campagnini et al. studied machine-learning methods for predicting functional recovery after stroke, supporting the idea that ML can be useful for recovery prediction. The literature in this area also highlights problems such as small datasets, limited external validation, and differences between datasets and variables used across studies.

2.1.5 Digital Exercise Applications

Davergne et al. studied digital applications that provide personalized exercise videos, demonstrating the usefulness of digital applications for exercise-based rehabilitation. However, such exercise applications alone do not necessarily provide continuous monitoring, recovery prediction, doctor supervision or treatment management.

2.1.6 Mobile Health Applications

Nussbaum et al. reviewed mobile health applications used in rehabilitation and found that mobile health can improve accessibility and patient engagement. However, effectiveness depends on usability, clinical integration and how well the application supports the overall rehabilitation process.

2.1.7 AI-based Exercise Assessment

Li et al. developed a CNN-based system for chronic stroke assessment and exercise recommendation, showing that AI can assist rehabilitation assessment. The approach, however, is focused on a specific condition and particular assessment tasks rather than acting as a general rehabilitation management platform.

2.1.8 Conversational AI in Healthcare

Li et al. reviewed AI conversational agents in healthcare and noted that conversational AI can support patients, but it should complement professional supervision rather than replace it — a principle that directly informs the design of the CAREX rehabilitation assistant.

2.1.9 ReWin Digital Rehabilitation Solution

A pair of studies on the ReWin digital rehabilitation solution demonstrate how structured digital rehabilitation systems can support patients with stroke-related disabilities in India. Their focus, however, is more condition-specific than the integrated, multi-condition approach that CAREX aims to provide.

2.2 COMPARISON AND DISCUSSION

The table below compares the major existing approaches to rehabilitation support against the CAREX approach.

Existing Approach	Main Capability	Major Limitation	CAREX Approach
In-clinic physiotherapy	Professional supervision	Periodic monitoring	Continuous digital monitoring
Paper / verbal instructions	Exercise guidance	No feedback	Interactive exercise support
Fitness applications	Activity tracking	Not rehabilitation-specific	Patient-specific rehabilitation
Telehealth	Remote consultation	Limited between-session monitoring	Continuous progress information

Existing Approach	Main Capability	Major Limitation	CAREX Approach
Exercise recommendation systems	Personalized exercises	Limited overall management	Personalized rehabilitation workflow
Recovery prediction models	Recovery analysis	Focused prediction task only	Prediction integrated with rehabilitation
AI assistants	Conversational support	Cannot replace professionals	Assistant + professional supervision
CAREX (proposed)	Integrated rehabilitation support	Requires further clinical validation	Personalized + AI + monitoring + doctor involvement

Table 2.1 Comparison of Existing Approaches with CAREX

2.3 CONCLUSION

The literature survey shows that artificial intelligence, telerehabilitation, personalized exercise recommendation, machine learning and conversational systems can each contribute to rehabilitation. However, many existing solutions focus on only one or two aspects of the recovery process. CAREX therefore combines personalized rehabilitation planning, exercise monitoring, progress tracking, recovery analysis, patient support and healthcare-professional supervision within a unified platform.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The CAREX platform is organized into five logical layers, described below.

Layer 1 — Patient Interface

Handles registration, patient profile, rehabilitation plan display, the exercise screen, health information, reminders and the AI assistant.

Layer 2 — Data Collection

Collects patient profile data, exercise information, progress data, health information and rehabilitation history.

Layer 3 — AI / Processing

Performs computer-vision-based exercise analysis, repetition counting, form evaluation, ML-based prediction / readiness analysis, and powers the AI assistant.

Layer 4 — Database / Cloud

Stores patient data, rehabilitation records, exercise session data, progress information and health information.

Layer 5 — Doctor Dashboard

Provides patient monitoring, progress analytics, reports and treatment-plan review / modification for healthcare professionals.

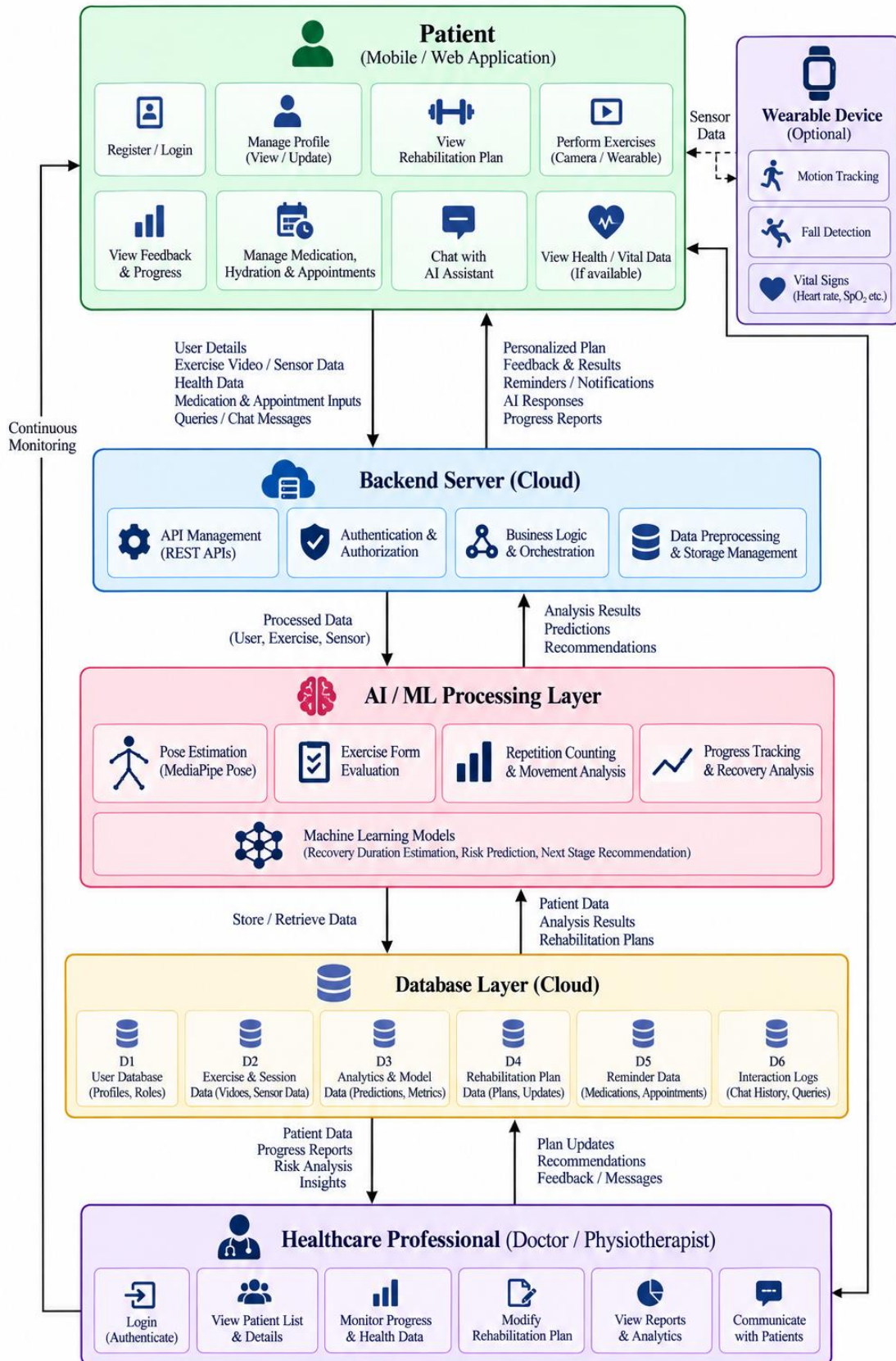


Fig. 3.1 CAREX System Architecture

Figure 3.1 illustrates the system architecture of CAREX and the interaction between its major components. The Patient interface collects profile, rehabilitation, exercise, and health-related information, while the optional wearable device provides sensor data such as movement and fall detection. The backend server manages authentication, business logic, data processing, and communication between the system components. The AI/ML layer performs pose estimation, exercise-form evaluation, repetition counting, progress tracking, and recovery analysis. The database stores patient, exercise, analytics, rehabilitation-plan, reminder, and interaction data. The Healthcare Professional dashboard allows professionals to monitor patient progress, review reports and insights, modify rehabilitation plans, and communicate with patients.

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

- Mobile / frontend application framework for the patient and doctor interfaces.
- Backend / API service layer for business logic and orchestration.
- Firebase for authentication, role-based access and real-time profile data.
- Computer-vision library (MediaPipe Pose) for pose landmark extraction.
- Machine-learning libraries for the XGBoost-based recovery/readiness model.
- Cloud / database services for persistent storage of rehabilitation records.

3.2.2 Hardware Requirements

Basic:

- Laptop / desktop for development and the doctor dashboard.
- Smartphone or device with a camera for exercise monitoring.
- Internet connectivity for cloud synchronization.

Optional:

- ESP32 microcontroller.
- MPU6050 6-axis IMU (accelerometer + gyroscope).

The ESP32 + MPU6050 combination forms an optional wearable extension, used only in situations where camera-based measurement may be affected by lighting or occlusion; it is not a mandatory part of the core system.

Category	Requirement	Purpose in CAREX
Frontend	Mobile / Web Application	Provides patient and healthcare-professional interfaces
Backend	Backend / API Service	Handles business logic and system communication
Authentication	Firebase Authentication	User authentication and role-based access
Computer Vision	MediaPipe Pose	Extracts body landmarks from exercise video
Machine Learning	XGBoost	Recovery and readiness analysis
Database	Cloud / Database Services	Stores rehabilitation and patient records
Development Device	Laptop / Desktop	Development and healthcare-professional dashboard
Patient Device	Smartphone / Camera-enabled Device	Captures exercise video
Optional Hardware	ESP32	Wearable sensing extension
Optional Sensor	MPU6050 IMU	Movement and fall detection

Table 3.1 — CAREX System Components

3.2.3 Data Requirements

Patient information: age, medical condition, injury / condition, recovery stage, treatment history.

Rehabilitation information: prescribed exercises, repetitions, sessions, exercise performance, progress.

Health information: relevant vitals, pain information and other patient-entered health data.

AI information: exercise metrics, recovery metrics, and readiness / recovery predictions.

Parameter	Type / Unit	Example
Age	Integer / Years	24
Repetitions	Integer / Count	10
Range of Motion	Float / Degrees	72.5°
Heart Rate	Integer / BPM	78
SpO ₂	Float / %	98
Pain Level	Integer / 0–10	3
Adherence	Float / %	85

Table 3.2 — CAREX Data Parameters

3.2.4 Security and Privacy

Because CAREX handles healthcare-related data, the design includes authentication, role-based access control, controlled patient/doctor access boundaries, secure storage and protection of patient information and privacy. Data security, privacy and clinical validation are treated as important considerations to be strengthened as the system matures.

3.3 SYSTEM DESIGN

3.3.1 Use Case Diagram

The two primary actors of the system are the Patient and the Doctor.

- **Patient:** Register / Login, Complete Profile, View Rehabilitation Plan, Perform Exercise, View Exercise Feedback, View Progress, Track Health, Manage Medication / Hydration, Ask AI Assistant.
- **Doctor/Health care Professional:** Login, View Patients, View Progress, View Reports, Review AI Analysis, Modify Treatment plan.

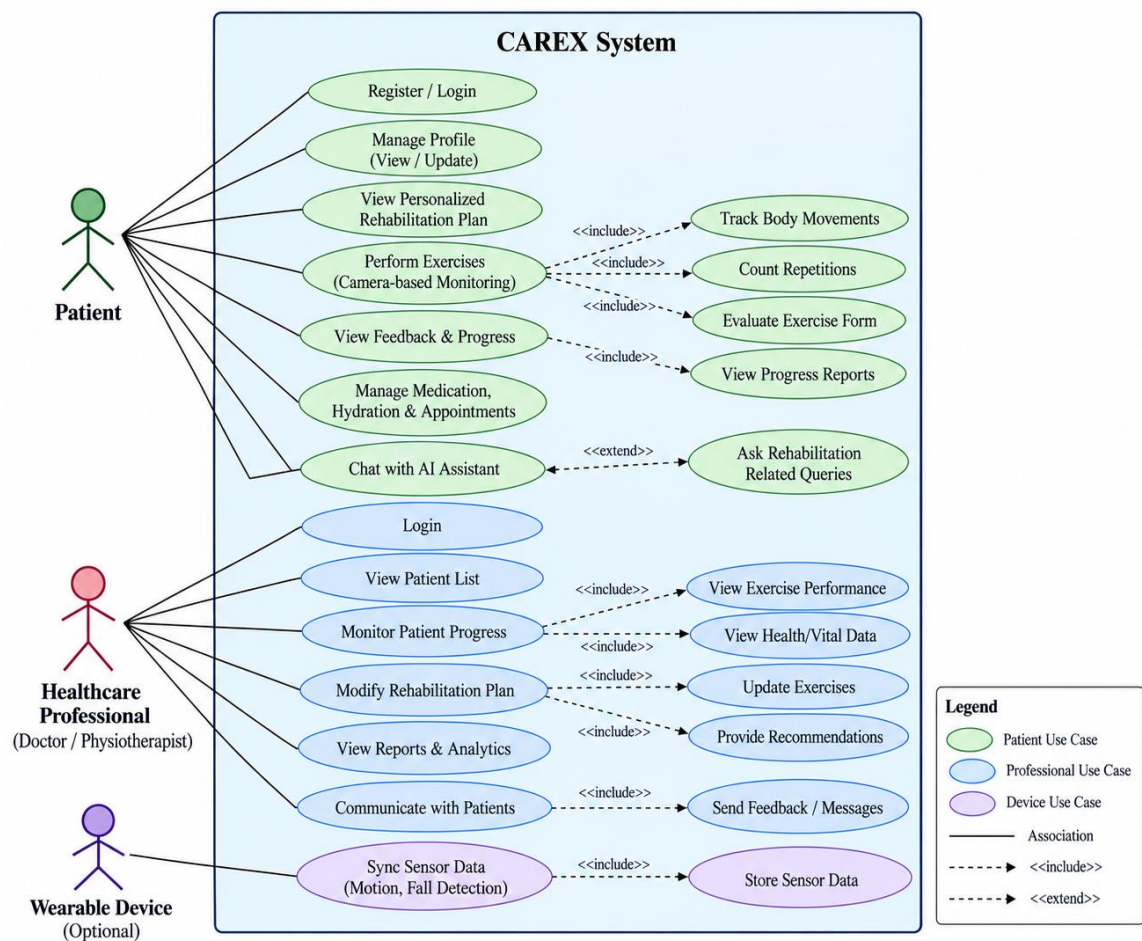


Fig. 3.2 Use Case Diagram

Figure 3.2 illustrates the major interactions between the actors and the CAREX system. The Patient can register and manage their profile, view the personalized rehabilitation plan, perform exercises with camera-based monitoring, view exercise feedback and progress, manage medication, hydration and appointments, and interact with

the AI assistant. The Healthcare Professional can log in, view patient details and progress, review reports and AI-based insights, modify rehabilitation plans, and communicate with patients. The optional Wearable Device provides sensor data such as movement and fall detection to the system. The diagram shows how these actors interact with the different functionalities of CAREX.

3.3.2 Activity Diagram

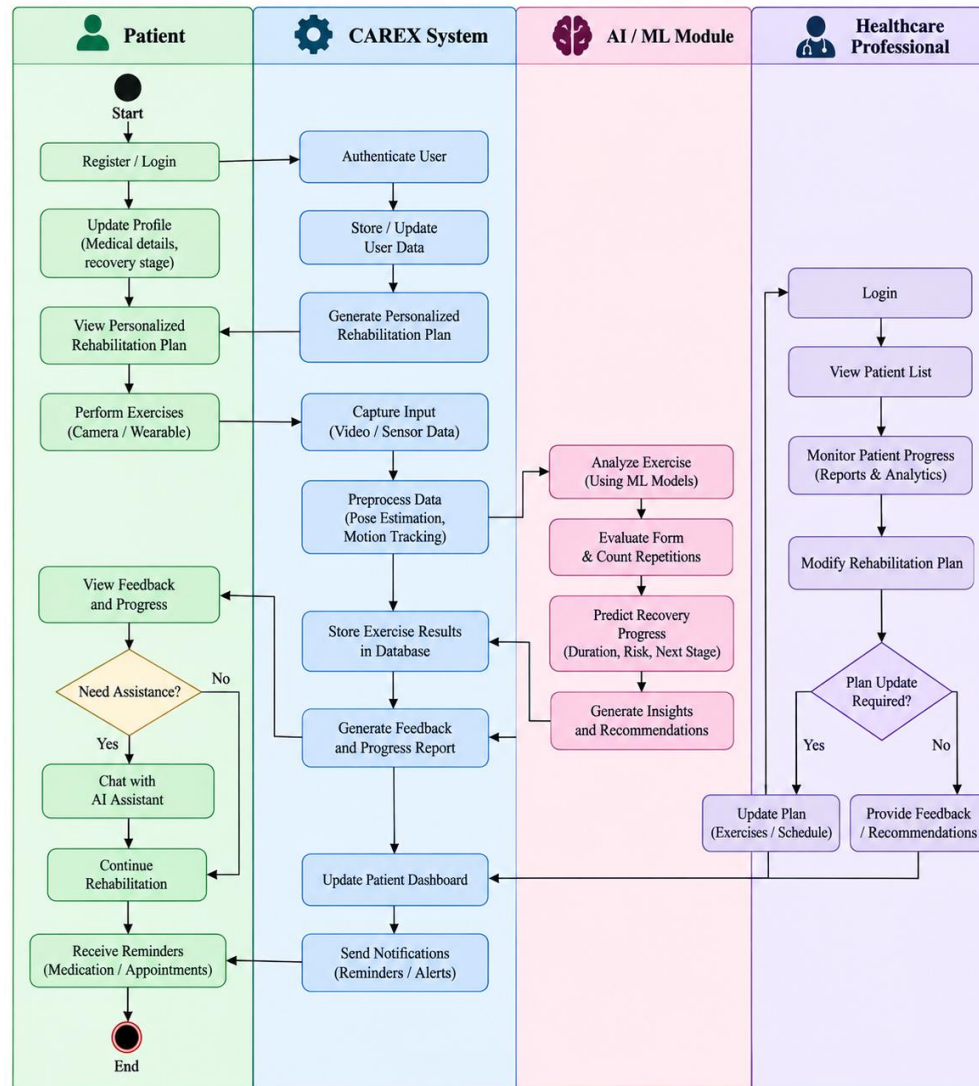


Fig. 3.3 Activity Diagram

Figure 3.3 represents the sequence of activities involved in the CAREX rehabilitation process. The process begins with patient registration and profile creation, followed by the generation of a personalized rehabilitation plan. The patient performs the prescribed exercises using camera or wearable-based monitoring, after which the system processes the collected data and analyzes exercise performance, repetitions, and recovery

progress. Feedback and progress information are provided to the patient, while healthcare professionals can monitor the results and modify the rehabilitation plan when required. The process also includes AI-assisted support and medication or appointment reminders to provide continuous assistance throughout rehabilitation.

3.3.3 Data Flow Diagram (Level 1)

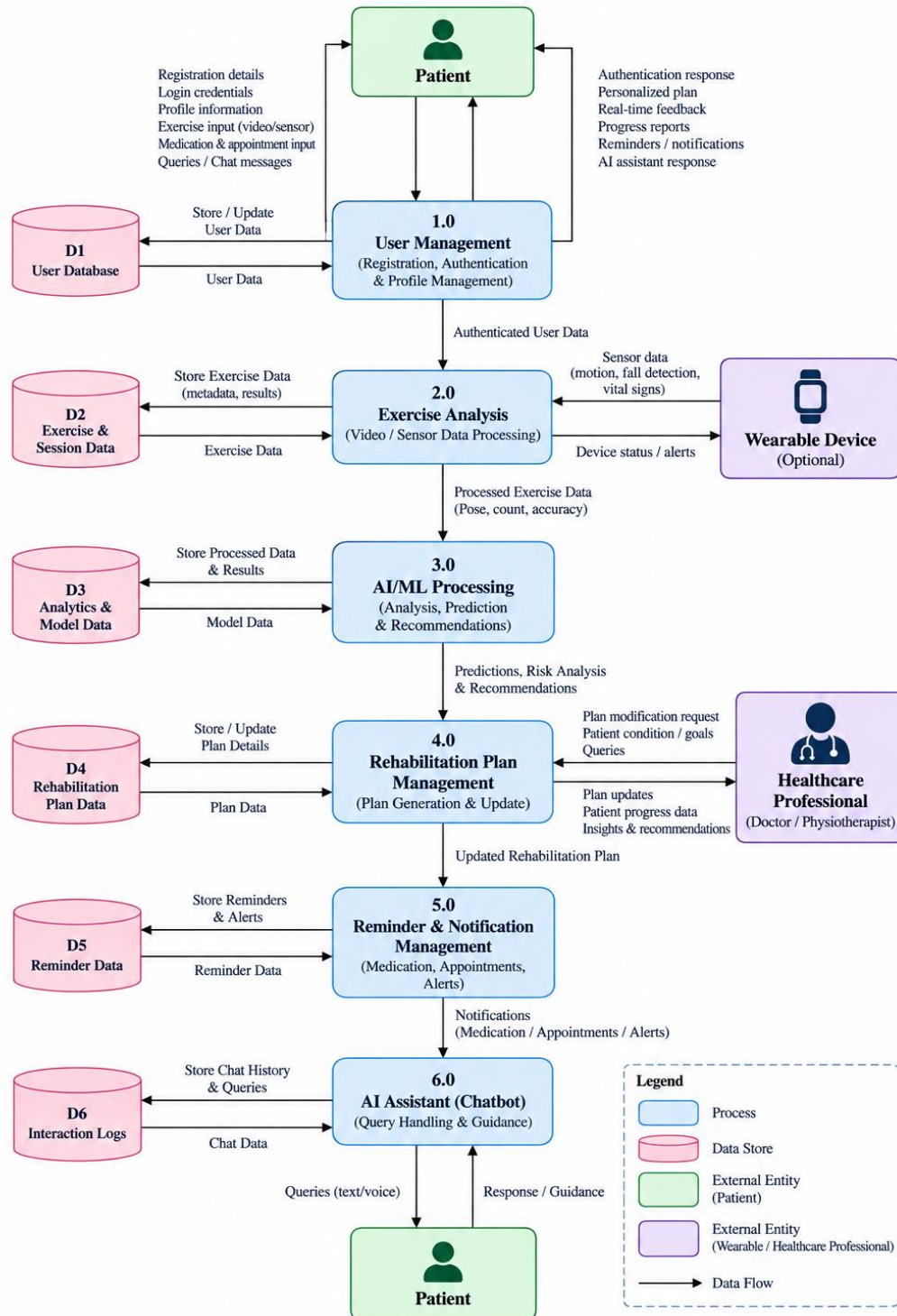


Fig. 3.4 Data Flow Diagram of the System

Figure 3.4 illustrates the Level 1 Data Flow Diagram of the CAREX system and shows how information moves between the patient, system processes, data stores, healthcare professional, and optional wearable device. The process begins with user registration, authentication, and profile management, after which patient and rehabilitation information is passed to the exercise and sensor data processing module. Exercise video and wearable sensor data are analyzed to obtain movement, repetition, form, and other relevant metrics. The AI/ML module uses the processed information to evaluate exercise performance, track progress, and support recovery analysis and recommendations. These results are used by the rehabilitation plan management module to generate or update personalized plans, while the reminder and notification module manages medication, appointment, and exercise-related reminders. The AI assistant handles patient queries and provides rehabilitation-related guidance. The healthcare professional can review patient progress, reports, and AI-generated insights and provide plan updates or recommendations. The respective data stores maintain user, exercise, analytics, rehabilitation-plan, reminder, and interaction information, allowing the system to support continuous and personalized rehabilitation.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

CAREX, which stands for Comprehensive Adaptive Rehabilitation Expert, is developed as an AI-powered personalized rehabilitation platform that integrates patient information, rehabilitation planning, exercise monitoring, progress tracking, recovery analysis and healthcare-professional supervision within a unified digital environment. The methodology followed in CAREX is designed around the continuous collection and analysis of patient-specific rehabilitation information so that the system can provide personalized support throughout the recovery process. The methodology begins with patient registration and profile creation, where information such as age, medical condition, injury or physical condition, recovery stage and treatment history is collected. This information is used to establish a personalized rehabilitation profile for the patient and forms the basis for recommending suitable rehabilitation activities.

After the patient profile is established, the rehabilitation plan is created according to the patient's condition and recovery stage. The patient can access the prescribed exercises through the application and perform them using a smartphone or camera-enabled device. During an exercise session, the computer-vision component captures the patient's body movement and uses MediaPipe Pose to identify relevant body landmarks. These landmarks are processed to understand movement patterns, estimate joint positions and angles, count repetitions and evaluate whether the exercise is being performed correctly. This allows CAREX to provide immediate exercise-related feedback rather than depending entirely on periodic physical consultations. The system architecture identifies exercise information and sensor information as important inputs to the processing layer, where movement, repetition and form-related metrics are obtained.

The processed exercise information is subsequently used for progress tracking and recovery analysis. CAREX considers indicators such as range of motion, exercise adherence, form compliance and pain trends to understand the patient's rehabilitation progress over time. The machine-learning component of the system is designed around XGBoost to support recovery and readiness analysis. The processed rehabilitation data can be used to estimate recovery-related outcomes, identify possible slow-recovery conditions and support recommendations regarding the next stage of rehabilitation. However, the AI

components are intended to provide analytical support rather than replace professional medical decisions. Healthcare professionals remain responsible for reviewing the patient's progress, evaluating AI-generated insights and modifying the rehabilitation plan whenever required.

The methodology also includes continuous patient support through reminders and an AI-powered rehabilitation assistant. Medication, hydration, appointment and exercise-related reminders are incorporated to improve adherence to the rehabilitation process, while the conversational assistant provides support for patient queries. At the healthcare-professional end, a dedicated dashboard allows doctors or physiotherapists to review patient information, exercise progress, reports and AI-generated insights and to update rehabilitation plans. The overall methodology therefore follows a continuous cycle of patient assessment, personalized planning, exercise execution, computer-vision-based analysis, progress tracking, recovery analysis and professional review. This approach allows CAREX to combine personalized rehabilitation with continuous digital monitoring while maintaining healthcare-professional involvement in treatment decisions.

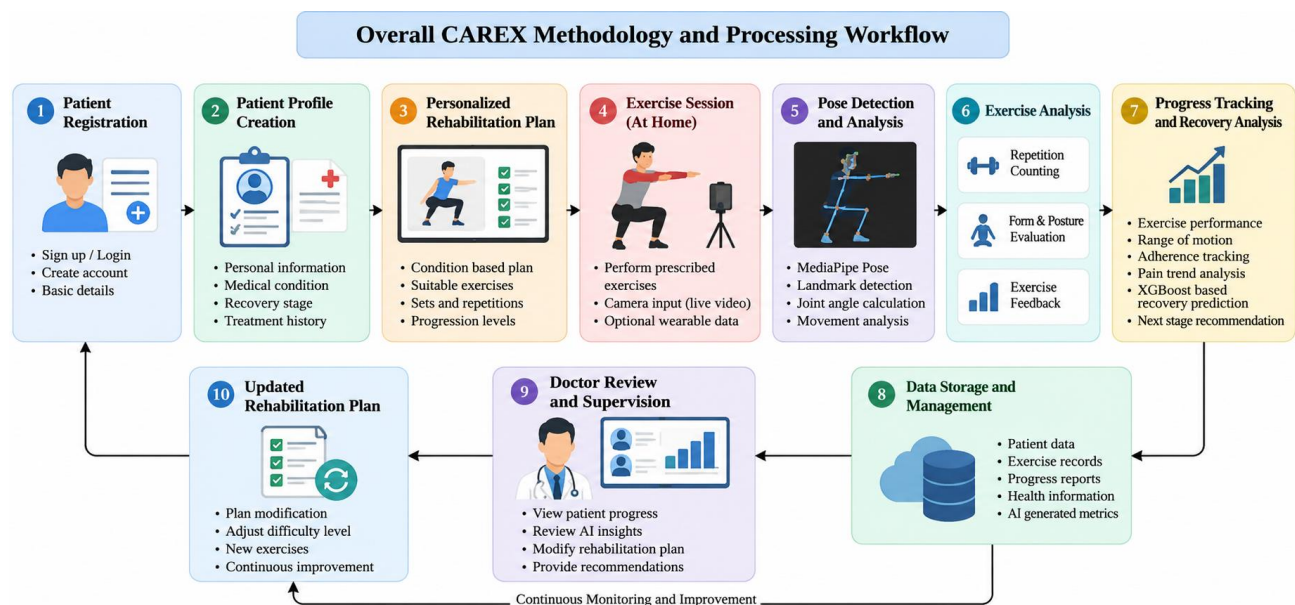


Figure 4.1: Overall CAREX Methodology and Processing Workflow

4.2 MODULES

The CAREX platform is organized into a set of functional modules that collectively implement the major activities involved in personalized rehabilitation. The modules are designed to separate patient management, rehabilitation activities, AI-based processing, professional supervision and continuous patient assistance. This modular organization

allows individual components to be developed and improved independently while maintaining communication between them through the backend and database layers. The architecture of CAREX consists of patient-facing functionality, data collection, AI and processing components, cloud or database services and a healthcare-professional dashboard.

4.2.1 DATA MANAGER PACKAGE

The Data Manager Package is responsible for collecting, organizing and maintaining the information required by the CAREX platform. The module manages patient-specific information including age, medical condition, injury or physical condition, recovery stage and treatment history. It also manages rehabilitation information such as prescribed exercises, repetitions, sessions and exercise performance, along with relevant health information, pain information and AI-generated rehabilitation metrics. The collected information is validated and stored so that it can be accessed by other components of the system whenever required. The data management layer acts as the foundation for personalization because the rehabilitation plan and subsequent analysis depend on the availability of accurate patient and rehabilitation information.

The Data Manager Package also maintains information generated during exercise sessions. When a patient performs an exercise, the system generates movement-related information such as repetitions, form-related measurements and other exercise metrics. These results are associated with the corresponding patient and exercise session and can subsequently be used for progress tracking and recovery analysis. The database and cloud layer stores patient information, exercise records, rehabilitation plans, progress information, reminders and interaction data. By maintaining these records in a structured manner, CAREX can preserve the patient's rehabilitation history and provide healthcare professionals with a consolidated view of the recovery process.

4.2.2 CORE PACKAGE

The Core Package represents the central processing component of CAREX and is responsible for implementing the primary rehabilitation workflow. It receives information from the patient and data-management components and coordinates the execution of rehabilitation plans, exercise monitoring, progress analysis and AI-assisted processing. The core component connects the patient interface with the computer-vision, machine-learning and database components so that information generated during rehabilitation sessions can

be processed and stored systematically. It therefore acts as the main coordination layer between the different functional components of the CAREX platform.

A major responsibility of the Core Package is exercise monitoring. During an exercise session, the patient's camera input is processed using MediaPipe Pose to identify body landmarks. The detected landmarks are used to analyse body movement and determine exercise-related parameters. Joint positions and movement patterns can be used to calculate relevant angles and identify movement transitions, which support automatic repetition counting and exercise-form evaluation. The results of this analysis are then made available to the patient as feedback and stored as part of the patient's rehabilitation history. CAREX specifically aims to evaluate exercise performance using computer vision, automatically count repetitions and identify incorrect posture or form.

The Core Package also coordinates progress tracking and recovery analysis. Exercise metrics collected over multiple sessions can be compared to identify changes in performance, adherence and range of motion. These processed features can then be supplied to the machine-learning component for recovery and readiness analysis. The architecture of CAREX places pose estimation, exercise-form evaluation, repetition counting, progress tracking and recovery analysis within the AI and processing layer, allowing these functions to work together as part of the rehabilitation workflow.

4.2.3 MODERATION PACKAGE

In CAREX, the moderation-related functionality is associated with maintaining the quality and appropriateness of information and recommendations presented to the patient. Since rehabilitation is a healthcare-related activity, the system must distinguish between analytical assistance provided by the AI components and actual clinical decisions made by healthcare professionals. The moderation layer therefore supports the controlled presentation of AI-generated information and ensures that recommendations are treated as supportive information rather than independent medical decisions.

The moderation process is particularly important for AI-generated recovery insights and patient interactions with the rehabilitation assistant. Information generated from exercise analysis and recovery prediction can be presented to the patient in a controlled manner, while more significant changes to the rehabilitation plan remain under the supervision of healthcare professionals. The CAREX design explicitly maintains professional involvement in treatment decisions, with healthcare professionals able to

review patient progress, examine AI-generated insights and modify rehabilitation plans when required. This approach provides a balance between automated assistance and professional supervision.

The moderation component can also support the handling of patient interactions with the AI assistant by ensuring that responses remain within the intended rehabilitation-support context. The assistant is designed to answer rehabilitation-related queries and provide day-to-day support, while clinical decisions remain with qualified healthcare professionals. This separation is important because CAREX is currently a proposed and developing system rather than a clinically validated replacement for professional rehabilitation services.

4.2.4 REINFORCEMENT PACKAGE

The Reinforcement Package in CAREX is associated with the continuous improvement of rehabilitation support using information collected from repeated patient interactions and exercise sessions. Rehabilitation is not a single-session activity; patient performance can change as recovery progresses. Therefore, the system maintains exercise history, progress information, adherence information and health-related observations so that subsequent rehabilitation decisions can be based on accumulated information rather than only the latest exercise session.

The package supports the adaptive nature of CAREX by allowing rehabilitation information to be reviewed over time. Exercise performance, range of motion, adherence and pain trends can contribute to understanding how the patient is progressing. These historical observations can be used by the AI and machine-learning components to support recovery and readiness analysis. The resulting insights can then be reviewed by healthcare professionals, who can modify the patient's rehabilitation plan when necessary. In this way, the system forms a continuous feedback cycle in which patient activity generates data, the system analyses the data, and healthcare professionals use the resulting information to guide future rehabilitation activities.

The adaptive process is especially relevant to the XGBoost-based recovery analysis proposed in CAREX. The machine-learning component is intended to analyse rehabilitation-related features and support recovery-duration estimation, slow-recovery risk identification and next-stage recommendations. Since the current Phase-I system

requires further validation using real clinical data, these outputs should be considered analytical support rather than clinically established predictions.

4.2.5 UI PACKAGE

The UI Package provides the interaction layer through which patients and healthcare professionals access the CAREX platform. The patient interface provides functions such as registration and login, profile management, rehabilitation-plan viewing, exercise execution, exercise feedback, progress monitoring, health tracking, medication and hydration management, appointment support and interaction with the AI rehabilitation assistant. The interface is designed to present the information required by patients in an accessible manner while allowing them to participate actively in their rehabilitation process.

The UI Package also includes the healthcare-professional interface. Through the doctor dashboard, healthcare professionals can log in, view patient information, monitor exercise progress, review reports and AI-generated insights, communicate with patients and modify rehabilitation plans. This creates two complementary interfaces within CAREX: one focused on patient participation and continuous rehabilitation support and another focused on professional monitoring and treatment-plan management. The user interface therefore acts as the final interaction layer between the underlying data, AI-processing components and the users of the system.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of CAREX follows the methodology and modular architecture described in the previous chapter. The implementation focuses on establishing the data-processing pipeline, exercise-monitoring functionality, AI-assisted analysis and user interaction components required for a personalized rehabilitation platform. The system processes patient information and rehabilitation data through different stages before presenting useful information to the patient or healthcare professional. The implementation is structured so that information generated during patient activity can be stored and reused for progress tracking and recovery analysis. The major implementation stages considered in this phase are data pre-processing, text processing where applicable, data representation, exercise analysis and AI-based rehabilitation analysis.

5.1.1 IMPLEMENTATION OF DATA PRE-PROCESSING

Data pre-processing is an important stage in CAREX because the quality of the information supplied to the analysis modules directly affects the usefulness of the resulting rehabilitation insights. The system receives different categories of information, including patient profile data, medical and rehabilitation information, exercise-session data, health information and AI-generated metrics. Before these data are used by subsequent modules, they need to be organized into a consistent structure. Patient information such as age, medical condition, recovery stage and treatment history is maintained as structured information, while exercise sessions are associated with the corresponding patient and prescribed exercise.

Exercise-monitoring data also undergo processing before being used for analysis. During a camera-based exercise session, the system receives video frames and extracts body landmarks using MediaPipe Pose. The detected landmarks provide the basis for calculating movement-related information and evaluating exercise performance. The processing stage can therefore convert raw camera observations into meaningful rehabilitation metrics such as movement information, repetition counts and form-related measurements. These metrics can subsequently be stored as part of the patient's exercise history and used for progress tracking. CAREX identifies exercise performance, repetition counting and form evaluation as key functions of its computer-vision component.

The pre-processed information is also used for progress and recovery analysis. Exercise performance can be considered along with adherence, range of motion and pain trends to understand changes over multiple rehabilitation sessions. The processed information is then made available to the machine-learning component for recovery and readiness analysis. By maintaining structured and consistent information throughout the pipeline, CAREX creates a foundation for personalized rehabilitation analysis and future model improvement.

Metric	Unit	Example Value
Session Duration	Minutes	12
Target Repetitions	Count	10
Completed Repetitions	Count	8
Correct Repetitions	Count	7
Average Joint Angle	Degrees (°)	68.4
Maximum Joint Angle	Degrees (°)	91.2
Minimum Joint Angle	Degrees (°)	43.7
Form Compliance	%	87.5

Table 5.1 — Exercise Session Measurements

5.1.2 IMPLEMENTATION OF DATA TOKENIZATION

Tokenization is normally used when textual information has to be converted into smaller units that can be processed by a natural-language-processing model. In CAREX, textual information can arise from patient queries, AI-assistant interactions and other text-based rehabilitation information. The tokenization stage can therefore be considered part of the processing pipeline for the conversational assistant when textual input is supplied to an underlying language model. The purpose of tokenization is to transform the user's

natural-language query into a machine-processable representation while preserving the meaning required for generating an appropriate response.

For example, when a patient enters a rehabilitation-related question, the input is first received by the AI-assistant interface and passed to the text-processing component. The text can then be divided into suitable tokens and converted into the representation expected by the underlying language model. The processed representation is subsequently supplied to the model for response generation. The resulting response is returned through the CAREX interface so that the patient can receive rehabilitation-related assistance. The Phase-I CAREX report establishes the presence of an AI-powered rehabilitation assistant for patient queries, but it does not specify a particular tokenizer or NLP architecture; therefore, the exact tokenizer and model configuration should be documented after the final implementation is fixed.

The tokenization stage is therefore considered a supporting component of the conversational functionality rather than the primary rehabilitation-analysis pipeline. The major numerical and visual data used by CAREX for exercise analysis are obtained from patient information, exercise sessions and pose landmarks rather than from textual tokenization. This distinction is important because CAREX combines different forms of information processing, with computer vision being used for exercise monitoring and language processing being relevant primarily to the conversational assistant.

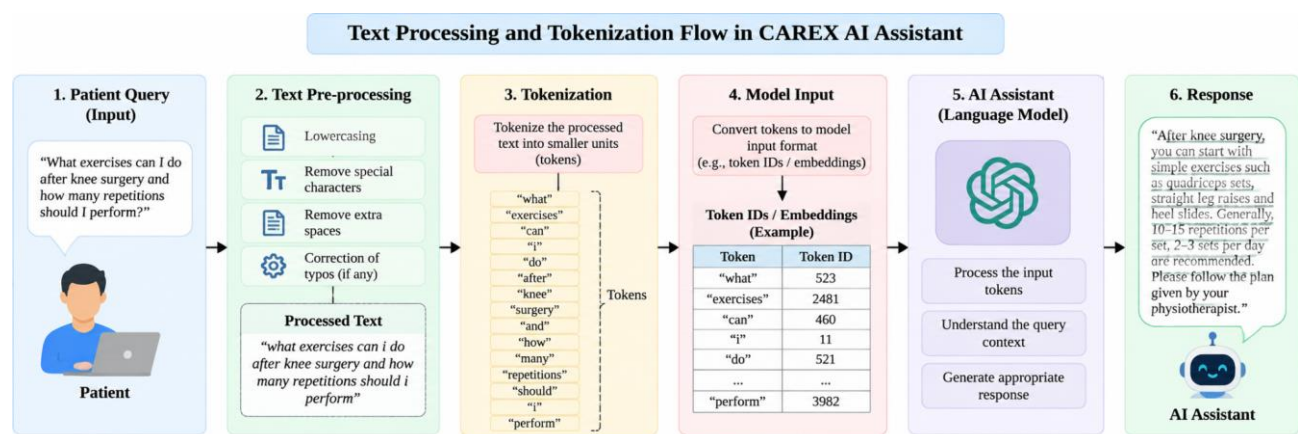


Figure 5.1: Text Processing and Tokenization Flow

5.1.3 IMPLEMENTATION OF DATA EMBEDDING

Data embedding refers to the conversion of processed information into numerical representations that can be used by machine-learning or artificial-intelligence models. Within CAREX, the requirement for numerical representation occurs in different

processing components. Exercise-related information obtained from pose landmarks can be represented through numerical features such as joint coordinates, joint angles, movement measurements and repetition-related values. These numerical features provide a structured representation of the patient's exercise performance and can be used for progress and recovery analysis.

The exercise-analysis pipeline begins with the extraction of body landmarks from camera frames using MediaPipe Pose. The landmark coordinates can be processed to derive movement-related features, including the relative positions of body joints and changes in joint angles during an exercise. These values form a numerical representation of the patient's movement and can be stored as exercise metrics. The resulting features can subsequently be combined with rehabilitation information such as adherence, pain trends and other relevant patient information for machine-learning-based recovery and readiness analysis. CAREX specifically proposes XGBoost for recovery and readiness analysis, making the preparation of meaningful numerical features an important part of the implementation pipeline.

For the conversational component, textual patient queries may also require numerical representation before being processed by the selected language model. However, the current Phase-I report does not identify a specific embedding model or embedding dimension, and therefore no particular embedding technique is claimed here. The exact embedding method can be documented in the subsequent implementation phase once the final AI-assistant architecture is established. The primary representation used by the current CAREX rehabilitation pipeline is the structured numerical representation of patient, exercise and movement-related information.

5.2 DISCUSSION

The implementation of CAREX demonstrates how artificial intelligence, computer vision, machine learning and digital healthcare components can be combined to create a unified rehabilitation-support platform. The major strength of the proposed implementation is that it does not treat exercise monitoring, progress tracking, recovery analysis and professional supervision as separate systems. Instead, information generated at one stage of rehabilitation can be stored and reused at later stages. Patient information is used to establish a personalized rehabilitation profile, exercise sessions generate performance

information, and the resulting metrics contribute to progress and recovery analysis. This creates a continuous information flow throughout the rehabilitation process.

The computer-vision component provides an important part of this workflow because it enables the system to analyse exercises performed outside a clinical environment. MediaPipe Pose is used to extract body landmarks from camera input, allowing the system to analyse movement, count repetitions and evaluate exercise form. This provides a basis for giving patients feedback during exercise sessions and collecting structured exercise-performance information for later analysis. The CAREX design also considers an optional ESP32 and MPU6050-based wearable extension for movement and fall-related sensing when camera-based measurement is affected by lighting or occlusion.

The progress-monitoring component extends the usefulness of individual exercise sessions by maintaining information over time. Indicators such as range of motion, adherence, form compliance and pain trends can be used to understand whether the patient is progressing through the rehabilitation process. The XGBoost-based recovery-analysis component is intended to use rehabilitation-related information to support recovery-duration estimation, slow-recovery risk identification and readiness analysis. These predictions are intended to provide analytical assistance and should not be interpreted as replacements for clinical assessment. The healthcare professional remains responsible for reviewing patient information and making treatment-related decisions.

The healthcare-professional dashboard provides an additional layer of supervision by allowing doctors or physiotherapists to review patient progress, reports and AI-generated insights. This is particularly important because rehabilitation is highly dependent on individual patient conditions and recovery patterns. By allowing professionals to modify rehabilitation plans based on the information collected by the system, CAREX follows a human-supervised AI approach rather than a fully autonomous treatment system. The AI components therefore act as decision-support mechanisms while clinical responsibility remains with the healthcare professional.

The current implementation also has limitations that must be considered when interpreting the results. CAREX is presently a Phase-I development and is not claimed to be a clinically validated product. The recovery and recommendation components require validation using real clinical rehabilitation datasets, while the exercise-monitoring component requires further testing across different body types, camera positions, lighting

conditions and exercise environments. The security and privacy mechanisms also need to be strengthened as the system progresses because CAREX handles health-related information. These limitations do not prevent the proposed architecture from serving as a foundation for further development, but they indicate that additional implementation, testing and clinical evaluation are required before the system can be considered for real-world healthcare deployment.

Overall, the CAREX implementation establishes a unified framework for personalized rehabilitation by combining patient-specific data, exercise monitoring, computer vision, machine-learning-based recovery analysis, reminders, conversational assistance and healthcare-professional supervision. The system is designed to reduce dependence on periodic monitoring by providing continuous digital support while retaining professional oversight. Future implementation can extend the existing framework through wearable integration, advanced computer vision, fall detection, multilingual assistance, improved predictive analytics and larger clinical datasets, allowing CAREX to evolve into a more comprehensive rehabilitation platform.

Session	ROM (°)	Pain Level (0–10)
1	54.2	6
2	58.6	5
3	63.1	4
4	67.8	3
5	71.5	2

Table 5.2 — CAREX Progress Indicators

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

CAREX brings together personalised exercise planning, continuous progress monitoring, recovery prediction, and clinician supervision within one platform, aiming to reduce the reliance conventional rehabilitation places on in-person visits and self-reported progress while keeping a clinician in charge of every treatment decision. The system integrates patient-specific information, rehabilitation activities, AI-assisted analysis, progress tracking, medication and appointment reminders, and a conversational assistant to provide continuous support during home-based rehabilitation.

Reviewing the literature shows that telerehabilitation, AI-based exercise recommendation, and machine-learning outcome prediction have each been studied on their own, with encouraging but methodologically limited results. CAREX combines these components into a single adaptive environment rather than proposing a new algorithm for any one of them. The architecture provides a foundation for personalised and data-driven rehabilitation while maintaining healthcare-professional involvement in the decision-making process.

The current system is a starting point rather than a clinically validated product. Further development is required to validate the recommendation and prediction models using real clinical data, evaluate the platform with patients and healthcare professionals, and strengthen the security and privacy mechanisms required for handling health-related information.

6.2 FUTURE ENHANCEMENTS

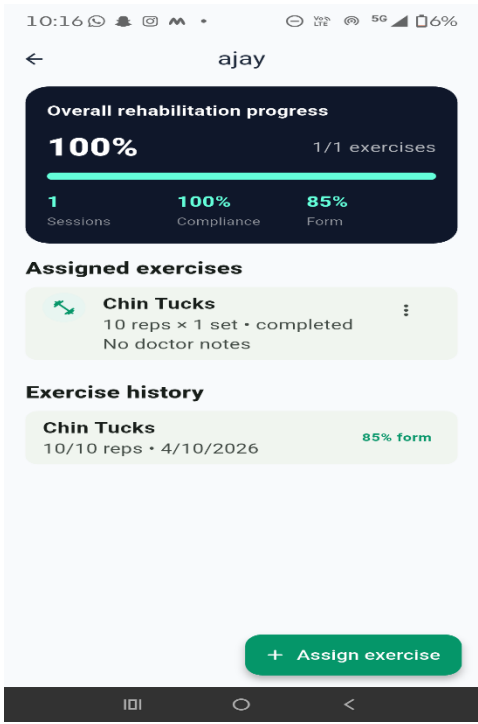
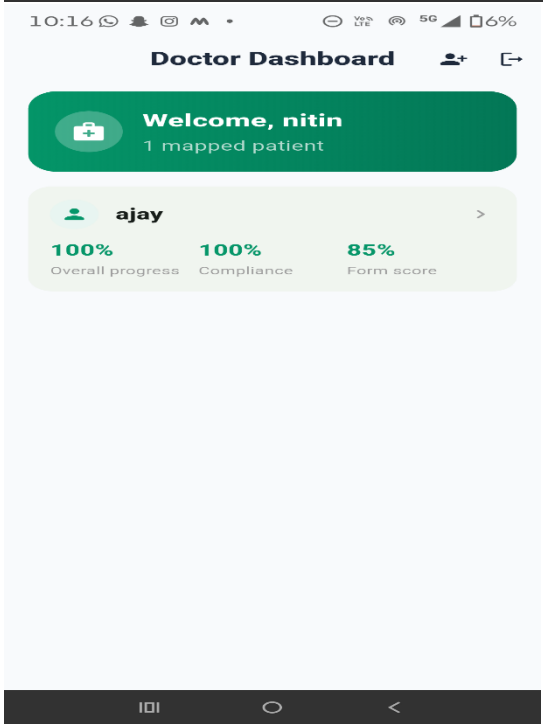
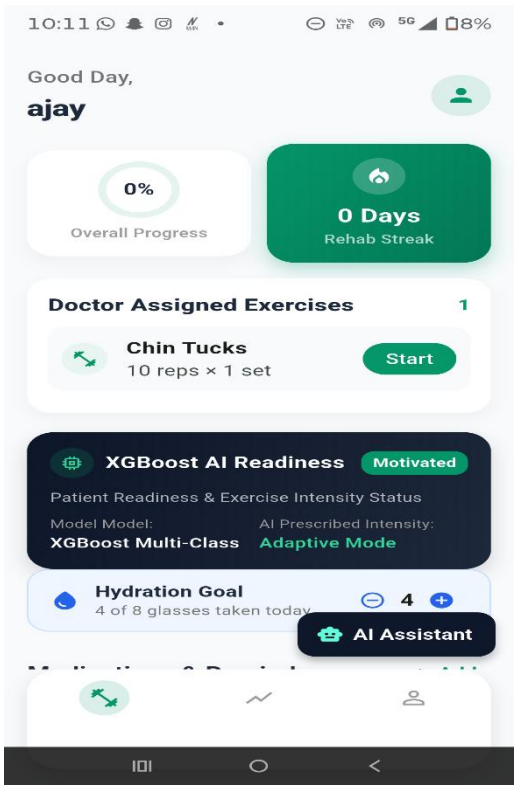
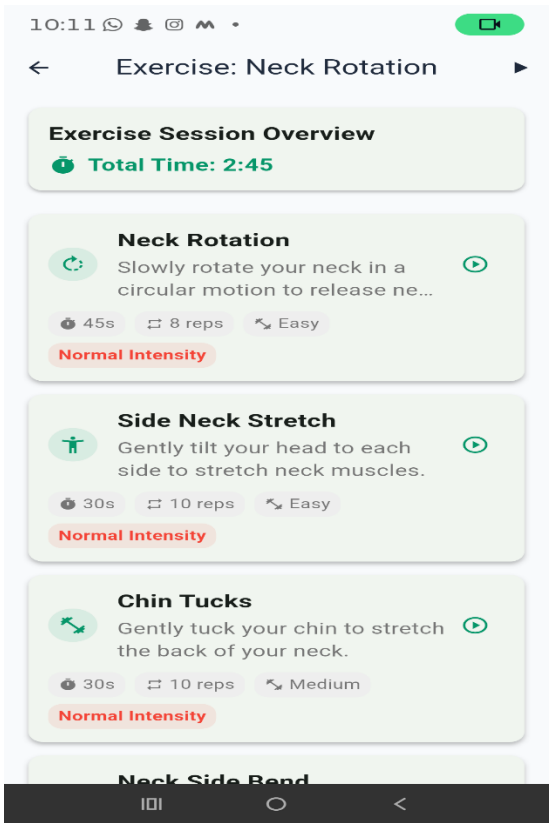
The future development of CAREX can focus on extending its capabilities for more accurate, accessible, and effective rehabilitation support. The major areas of future enhancement include:

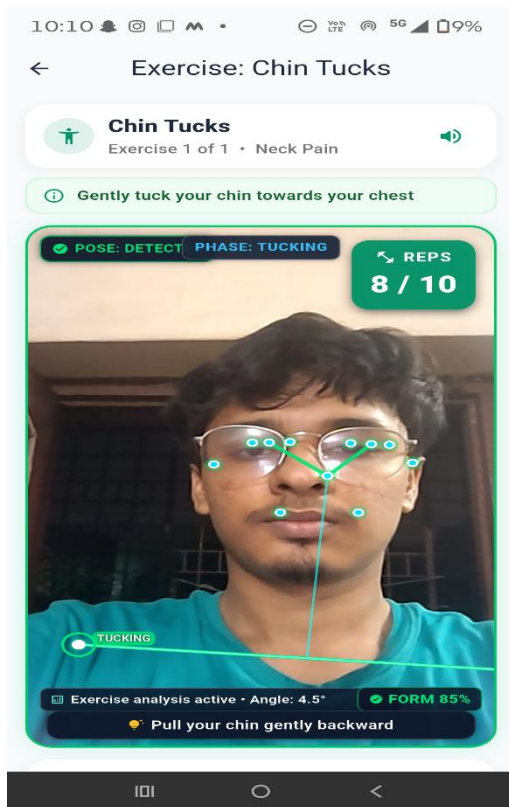
- **IoT-Based Wearable Integration:** Integrate an ESP32 and MPU6050-based wearable device to capture real-time movement and motion data during rehabilitation exercises.

- **Real-Time Exercise Monitoring:** Combine wearable sensor data with computer vision to improve exercise recognition, movement tracking, and posture assessment.
- **Fall and Abnormal-Movement Detection:** Detect sudden falls or abnormal movement patterns and trigger SOS alerts to caregivers or healthcare professionals.
- **Advanced Computer Vision:** Enhance pose estimation with joint-angle, range-of-motion, repetition, and exercise-correctness analysis for more detailed rehabilitation monitoring.
- **Doctor/Clinician Portal:** Enable healthcare professionals to remotely monitor patient progress, review AI-generated insights, and update rehabilitation plans.
- **Voice and Multilingual Assistant:** Provide hands-free exercise instructions, real-time feedback, and multilingual guidance to improve accessibility.
- **Predictive Recovery Analytics:** Improve machine-learning models using larger rehabilitation datasets to predict recovery progress, potential risks, and required changes in exercise intensity.
- **Gamification and Patient Engagement:** Introduce recovery milestones, badges, goals, and progress-based rewards to encourage consistent participation in rehabilitation.

These enhancements can help CAREX evolve into a more comprehensive rehabilitation platform while continuing to maintain healthcare-professional involvement in patient care.

APPENDIX





CAREX: Comprehensive Adaptive Rehabilitation Expert – A Proposed AI-Powered Personalized Rehabilitation Platform

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Abstract

Effective rehabilitation following surgery, fracture, stroke, or sports injury occurs when rehabilitation is continuous, well-monitored, and adjusted according to the patient's recovery progress. In actual practice, however, continuous personalized care is hard to deliver: it is difficult for clinicians to see patients often enough to make accurate assessments and patients often need to report their own progress. The same standard exercise regimen is usually prescribed to patients regardless of their individual recovery progress. This article presents CAREX (Comprehensive Adaptive Rehabilitation Expert), an AI-based rehabilitation program aimed at overcoming this shortcoming. CAREX creates a rehabilitation profile of the patient based on the patient's age, medical condition, stage of recovery, and medical history, and combines this profile with data on the extent of exercise and the rate of progress made in order to develop a personalized exercise plan. With every new progress report, CAREX reads the patient recovery situation once again so that it can re-evaluate its recommendations and offer estimates of the expected recovery progress. A doctor dashboard serves as a communication point that allows medical professionals to monitor and modify the treatment plan. Taken together, these components are intended to make rehabilitation more consistent for the patient and easier to monitor for the clinician, without removing the clinician from the decision-making process.

Keywords: Adaptive rehabilitation; Artificial intelligence; Digital health; Recovery prediction; Telerehabilitation

1. INTRODUCTION

Rehabilitation is often the deciding factor in how well a patient recovers from an injury, a surgical procedure, or a neurological event such as a stroke. Good outcomes generally require regular exercise to be carried out by the patient, along with periodic reviews with a clinician. The way this has traditionally been delivered, however, involves in-person supervision plus the patient's honest and accurate reporting of their progress, and it is genuinely difficult for a clinician to know where they are between visits. Some patients find it challenging to keep up with their exercises, medications, and appointments amongst their other

commitments. Artificial intelligence and digital health solutions can help to close this gap. Instead of prescribing the same rehabilitation pathway to all patients with the same diagnosis, a system which has an understanding of the patient's individual condition, age, recovery status, and treatment history can personalise its recommendations to that person and update itself as the recovery progresses. It can also store a complete history of the patient's progress and display it to the clinician in a way that aids more informed decision-making, rather than relying on what a patient can remember at a short consultation.

CAREX (Comprehensive Adaptive Rehabilitation

Expert) is proposed in this proposal as a platform to connect patients and clinicians through the same digital interface. It generates a personalised schedule of exercises from patient information, tracks the rehabilitation progress continuously, and provides an outlook on the recovery that could be utilised by a clinician to aid in their own judgement. A dedicated dashboard for doctors allows the clinician to assess performance, review analytics, and modify a treatment plan from a remote location, reducing the amount of patient progress to be tracked manually.

Medication and appointment reminders, structured health reports, and an AI-based virtual assistant complement the platform. Together, they are intended to make life easier for the rehabilitation journey for the patient and for oversight to be provided by the clinician, while still remaining in the control of the clinician to make treatment decisions.

1.1. Related Work

Telerehabilitation has been investigated as an alternative to traditional forms of personal therapy for some time now. Laver et al. (2020) conducted a review of telerehabilitation services for post-stroke patients, evaluating effectiveness relative to traditional institutional therapy. Their analysis demonstrated that remote rehabilitation produced comparable results to conventional therapy in a number of considered settings, thus proving to be an effective substitute in terms of reducing the frequency of a patient's visits to a healthcare facility. However, none of the reviewed studies attempted to tackle the issue of personalized recommendations, focusing primarily on the logistics of delivering therapy from a remote location.

An emerging body of research has devoted its attention to the role of artificial intelligence in physical rehabilitation as a tool for enabling personalized recommendations. Sumner et al. (2023) performed a systematic review of AI-enabled physical rehabilitation technologies. Their work

provides an overview of the current state-of-the-art in the field, outlining the various methods, applications, and tools. In their review, Sumner et al. note the general ability of AI technologies to facilitate accessibility in physical rehabilitation and enable remote supervision while identifying a number of barriers stemming from technology readiness, technical literacy, fatigue, and the absence of clinical evaluation of such technologies.

The issue of personalized recommendations has been approached from multiple angles in the literature. Kim et al. (2021) designed an AI-based recommendation system that employs a graph neural network to recommend personalized therapeutic exercises based on the individual patient's characteristics. The focus of their research was on the recommendation algorithm itself and the architecture of the graph, without considering broader aspects of therapeutic management, such as tracking of therapeutic progress, reminders, or post-therapeutic analysis and evaluation in conjunction with medical professionals.

On the other hand, Campagnini et al. (2022) performed a systematic review of the role of machine learning in outcome prediction in post-stroke rehabilitation. Their review identified a variety of studies employing ML technologies to predict patient outcomes based on various clinical and demographic features, utilizing both conventional statistical models and complex machine learning algorithms. The review highlighted the general limitations of the ML-based approach in this domain, including the lack of extensive validation data and consistency between studies, small sample sizes, and issues with generalizability stemming from inconsistencies between input variables and clinical predictors.

The focus of Davergne et al.'s (2023) review was on digital interventions providing personalized video exercises, which analyzed the effect of such applications on adherence and outcomes through a meta-analysis. Their work demonstrated the general benefits of such personalized digital interventions

while highlighting heterogeneity between the included studies. Similarly, Nussbaum et al. (2019)’s review focused on the role of mobile technologies in post-stroke rehabilitation, analyzing the current applications and evaluating their effectiveness. Their review highlighted the present limitations of mobile technologies in post-stroke rehabilitation, focusing on the lack of comprehensive tools, which are typically limited to providing a single function, such as tracking or training, as opposed to integrated solutions incorporating assessment, planning, prediction, and clinical management, such as CAREX.

Kamalakannan et al. (2022a; 2022b) describe the development of a mobile application for upper limb rehabilitation, entitled ReWin, and evaluate its feasibility and acceptability among target users in India. Their work demonstrates that such applications are generally met with positive feedback from patients and caregivers. Gokul Prasath et al. (2066) performed a survey among occupational therapists in India regarding their use of mobile applications for neurological rehabilitation. Their findings identified the general level of adoption among medical professionals as low due to a combination of factors, including lack of awareness and compatibility with the existing practices. Finally, Badhal et al. (2026)’s work focuses on developing an ethical AI framework for neurological rehabilitation, postulating that AI technologies should be utilized to complement rather than replace the role of clinicians, which is in alignment with CAREX’s approach to the doctor dashboard, which focuses on physician assessment rather than attempting to replace it.

2. METHOD

CAREX is structured with five interconnected elements namely: a patient application, a backend service layer, a rehab database, an artificial intelligence system, and a health professionals’ dashboard. The platform is designed to be used by patients undergoing rehabilitation after surgery,

fracture, an injury related to sports, or stroke, as well as other related musculoskeletal conditions. Unlike traditional patient care, where each new consultation is seen as a discrete event, CAREX uses a continuous rehabilitation profile for each patient and regularly updates it with information as it becomes available.



FIGURE 1. Architecture of the CAREX platform

2.1 System Architecture

The patient application collects information about the patient's status in addition to processing info and delivers them to CAREX backend which effectively manages data input and output to the signed database which provides data for the treatment and rehabilitation of patients. The backend user will get answers and information on their treatment in the AI system, including notifications about the possibilities of receiving the recommendation and predictions. The physician's console works parallel with this process and is receiving information from the backend and allows storing a new approach to treatment in the system.

2.2 Patient Data Preparation

A patient's overall state can be represented using two complementary components: relatively stable profile characteristics and rehabilitation-related information that changes over time. This representation can be expressed as

$$X = [Xp, Xr] \quad (1)$$

where Xp represents the patient's profile-related attributes, including medical condition, age, recovery stage, and treatment history. Xr represents rehabilitation data gathered during the recovery process, such as completed exercises and recorded progress.

Since the dataset contains both numerical and categorical attributes, the data must be preprocessed before being provided to the AI model. Categorical values are transformed into a machine-readable representation, while numerical attributes are scaled to a common range. This helps prevent features with larger numerical values from having an unintended influence on the model.

The normalization of a numerical value x is performed using

$$x' = (x - xmin) / (xmax - xmin) \quad (2)$$

Historical rehabilitation records are also summarised into progress-related features that describe a patient's recent activity and recovery status, and it is this resulting feature vector that feeds into the profile-analysis and recovery-analysis stages that follow.

2.3 AI-Based Representation and Personalised Recommendation

The AI engine first builds an internal representation of the patient's profile, then a separate representation from the patient's rehabilitation history, and finally combines the two before generating any recommendation. This is expressed as

$$Zp = f(Xp), \quad Zr = g(Rt), \quad Z = F(Zp, Zr) \quad (3)$$

where Rt is the rehabilitation information accumulated up to time t . While Zp identifies the medical context of a patient, Zr reflects the treatment response of the patient; this information can subsequently be processed by the downstream modules to reflect both aspects at once in their operation instead of viewing the diagnosis and the actual progress as independent occurrences.

The recommendation engine then scores each candidate exercise ei according to its fit with the current patient state:

$$S(ei) = w1Ci + w2Ri + w3Hi + w4Pi \quad (4)$$

where Ci , Ri , Hi , and Pi indicate, respectively, the appropriateness of the exercise according to the patient's condition, the appropriateness for the current recovery phase, the compatibility with treatment history, and the appropriateness of observed outcomes; and weights $w1$ – $w4$ show the relative importance of each criterion. Table 1 shows how each of these criteria is evaluated. The resulting scores give a ranking of exercises; only the highest ranked ones are selected to create the plan. The fact that the score is calculated in accordance with the features of the patient rather than only the diagnosis means that two patients diagnosed with the same condition receive different plans.

Table 1. Factors used in the exercise suitability score

Factor	What it captures
Ci	Compatibility with the patient's medical condition
Ri	Suitability for the current recovery stage
Hi	Consistency with treatment history
Pi	Fit with the patient's observed progress

2.4 Adaptive Feedback and Recovery Prediction

As the patient progresses through the recommended plan, new progress data flows back into the backend and into subsequent recovery analysis. The initial plan is never treated as fixed – later recommendations are generated from the updated patient state. A recovery prediction component estimates the likely path of recovery:

$$\hat{Y} = P(Zp, Zr) \quad (5)$$

This prediction should be used as an adjunct to monitoring and not as a sole basis for treatment. It is surfaced through the doctor dashboard where a clinician can accept, adjust or override the AI-generated plan based on their own judgment – a human-in-the-loop step of analysis, recommendation, professional review and an approved or modified plan. Supporting services, which are alongside this loop, aim to improve adherence and accessibility without replacing clinical supervision and include medication and appointment reminders, structured health reports and a conversational assistant for rehabilitation-related queries.

3. RESULTS AND DISCUSSION

3.1 Results

The results discussed in this section are the functional capabilities derived from the design specified in Section 2, and not measured clinical outcomes, since CAREX is presented here as a proposed system architecture and not a clinically deployed product. Structurally the design achieves three things that are usually only addressed individually in the literature reviewed. Finally, by separating profile information (X_p) from recovery information (X_r) and then recombining them before recommendation, the system can distinguish a patient's

initial clinical context from their actual response to treatment instead of reducing both to a static diagnosis-to-exercise mapping. Second, the exercise score $S(e_i)$ is built on four independent factors, so the resulting plan changes in a traceable way when any of these factors changes. This is not the case for systems recommending only from diagnosis. Third, routing each plan and prediction generated by AI through the doctor dashboard before it reaches the patient keeps a clinician in the loop at each cycle, rather than only at scheduled visits—the same human-in-the-loop pattern that Badhal et al. (2026) argue for at a conceptual level.

The resulting operating cycle is summarized in Figure 2: patient data is analyzed by the AI engine into a personalized plan, tracked through progress monitoring, fed back into recovery analysis and prediction, reviewed by a clinician, and finally fed back to the patient as an updated plan — at which point the cycle repeats using the newly observed progress.

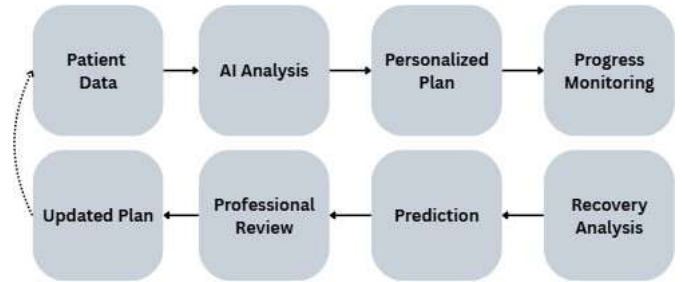


FIGURE 2. The continuous feedback cycle underlying CAREX's operation

3.2 Discussion

The main contribution of this design is not so much the single algorithm, but rather the way in which it integrates personalized recommendation, recovery prediction, clinician supervision and adherence support into a single profile per patient, separately addressed in Kim et al. (2021) , Campagnini et al. (2022) and Nussbaum et al. (2019) . But whether this actually improves outcomes in practice is a separate question from whether the architecture is coherent, and it can only be answered with real deployment.

This takes us directly to a couple of limitations. There the weights w_1 – w_4 in the recommendation score

are treated as design parameters. In the absence of clinical data to fit or validate against, there is a real danger of a plan that looks reasonable on paper but is not actually well calibrated for a given condition, which echoes the validation concerns raised by Campagnini et al. (2022) about recovery-prediction models more generally. Technology literacy and user fatigue, as Sumner et al. (2023) also observe, limit the day-to-day use of AI-based rehabilitation tools, and CAREX is not immune to that: a patient who finds the application confusing or tiresome to use is unlikely to benefit from how well the recommendation engine performs behind the scenes. Gokul Prasath et al. (2026) describe a similar pattern among occupational therapists in India, where the uptake of mobile rehabilitation tools is heavily dependent on how well a tool fits into existing clinical workflow, which is as much a design and change-management question as a technical one.

The present architecture makes the wearable part optional, and focuses on software and AI workflow. Extending it to include continuous sensor data will narrow the recovery representation Z_r , but brings its own reliability and battery-life constraints, which is why it is not in the present scope. To summarise, the design can be considered a base, where certain parts – the recommendation weights, the prediction model, and the assistant's dialogue behaviour – would need to be individually clinically evaluated and validated before the platform could be used without supervision.

CONCLUSION

CAREX offers personalised exercise planning, continuous progress tracking, recovery forecasting and clinician oversight in a single platform, aiming to decrease the reliance of conventional rehabilitation on face-to-face visits and self-reported progress, while maintaining clinician control over every treatment decision. A review of the literature reveals that telerehabilitation, AI-based exercise recommendation, and machine-learning outcome prediction have been studied separately with promising, but methodologically limited results (Laver et al., 2020; Sumner et al., 2023; Campagnini et al., 2022). CAREX's contribution is to combine these components in a single adaptive

environment, rather than proposing a new algorithm for any one of them. The architecture described here is a first step and not a finished product. The next steps are to fit and validate the recommendation weights and prediction model against real clinical data, run the platform through a real-world evaluation with patients and clinicians, and bolster the security and privacy protections that any system handling health data needs before it can be used at scale.

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Paper ID: 2609030

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Paper Title: CAREX: Comprehensive Adaptive Rehabilitation Expert

Abstract: Rehabilitation requires regular exercise, progress monitoring, and timely adjustment of treatment according to a patient's recovery. However, continuous supervision between clinical consultations can be difficult. This paper proposes CAREX (Comprehensive Adaptive Rehabilitation Expert), an AI-powered platform for personalized rehabilitation support. The proposed system provides patient profiling, adaptive exercise recommendations, recovery progress estimation, and continuous monitoring to support individual rehabilitation needs. It also includes a doctor dashboard for monitoring patient progress, reminders and reports for better adherence, and an AI-based virtual assistant for guidance and support. The platform is designed to improve accessibility, personalization, and continuity of rehabilitation care. The proposed architecture is intended for further validation with clinical data and expert evaluation before real-world deployment.

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BusMind: An Emotion-Aware AI Study Companion for Bus Commuters

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ABSTRACT:

BusMind is a mobile AI study companion designed for students who spend a significant amount of time travelling by bus. Instead of treating commuting as unused time, the system attempts to convert short and fragmented travel intervals into manageable learning sessions while also considering the learner's current context. The application integrates facial-expression-based emotion detection, mood-aware music recommendation, smart flashcards, voice-based note summarization, posture monitoring, learning analytics, and graph-based study-path organization within a Flutter application. The system uses a Vision Transformer-based facial-expression model to obtain a broad expression category, which is subsequently converted into an application-level mood state. This mood information is combined with user preferences and session context before generating recommendations. The learning component uses locally stored flashcards and a spaced-review mechanism to prioritize cards that require attention. Voice input is converted into text using device-supported speech recognition and can subsequently be summarized into concise study material. Accelerometer and gyroscope readings are processed together to identify persistent device-orientation patterns and provide posture-awareness reminders. The application follows a modular architecture in which authentication, mood detection, music, flashcards, voice interaction, posture monitoring, and analytics are managed through separate application providers. Local persistence through Hive and Shared references allows several learning functions to remain available when network connectivity is limited. The system also incorporates BFS, DFS, and A* algorithms to organize topic dependencies and recommend feasible learning paths. The contribution of BusMind is primarily architectural and functional. The proposed system demonstrates how multiple lightweight AI, learning, and mobile-sensing capabilities can be combined into a single commute-oriented learning workflow. It does not claim clinical, psychological, or statistically validated educational benefits without controlled user studies.

Keywords: Mobile learning; Emotion-aware computing; Personalized learning; Facial expression recognition; Spaced repetition; Voice summarization; Posture monitoring; Learning analytics; Flutter

I. INTRODUCTION

Mobile devices have become an important part of the daily learning environment of university students. Their portability allows learning activities to continue outside classrooms, laboratories, and homes, making mobile learning particularly suitable for situations in which students have short periods of otherwise unused time. A systematic review by Crompton and Burke reported extensive use of mobile learning in higher education and highlighted the potential of mobile devices to extend learning beyond conventional educational settings [1]. However, the availability of a mobile device alone does not ensure that fragmented periods such as daily bus journeys become productive learning sessions.

A bus journey differs significantly from a conventional study environment. The learner may have only a few minutes available, the surrounding environment may be noisy or distracting, internet connectivity may

fluctuate, and the learner's attention and energy may vary throughout the journey. A learning system designed for such an environment therefore needs to be flexible rather than assuming that every session should follow the same learning pattern.

BusMind is proposed as a context-aware mobile study companion that attempts to address this problem. Instead of functioning as a conventional collection of learning tools, the system combines several services within a single workflow. The learner can review flashcards, listen to recommended music, capture study notes using voice, monitor posture, and inspect learning progress without moving between multiple applications.

An important characteristic of BusMind is that it treats the learner's context as an input to the system. Facial-expression analysis provides an optional contextual signal, while available time, learning history, topic priority, and user preference provide additional information. This follows the broader concept of context-aware recommendation, where recommendations are generated by considering the situation in which the user is interacting with the system [2].

The objective is not to claim that a facial expression completely describes a learner's emotional state. Instead, the detected expression is converted into a coarse application-level context that may influence recommendations. This distinction is important because emotion recognition remains sensitive to factors such as lighting, facial orientation, dataset characteristics, and individual differences.

The learning component of BusMind is based on retrieval and spacing principles. Retrieval practice requires learners to actively recall information, while spacing distributes learning opportunities over time. Research has consistently identified these strategies as useful mechanisms for supporting long-term learning and retention [3], [4]. BusMind applies these principles through an adaptive flashcard scheduler that gives greater attention to cards with lower recent recall or longer review intervals.

The major contribution of the proposed system is therefore the integration of these capabilities around a specific real-world context: learning during bus travel. Rather than introducing another isolated flashcard, recommendation, or sensing application, BusMind explores how these components can cooperate within a lightweight mobile architecture.

II. RELATED WORK

A. Mobile Learning

Mobile learning enables educational interaction through portable devices across different locations and contexts. Crompton and Burke's systematic review of mobile learning in higher education identified learning achievement, learner perception, pedagogy, and other dimensions as important research areas [1]. Their findings demonstrate that mobile devices can extend learning opportunities beyond formal classrooms.

However, many existing mobile-learning applications primarily focus on content access. BusMind extends this concept by considering the conditions under which learning occurs. The proposed system is designed for short, repeated, and potentially interrupted sessions,

making activity selection and persistence important components of the architecture.

B. Context-Aware Recommendation

Traditional recommendation systems commonly rely on user preferences and historical interactions. Context-aware recommender systems additionally incorporate information such as time, location, activity, or environmental conditions [2]. This principle is particularly relevant to BusMind because the usefulness of an activity can depend on the remaining commute time and current learning state.

For example, a learner with ten minutes remaining may benefit from a small flashcard session, whereas a learner with a longer journey may choose voice-based note creation or a deeper study activity. The proposed recommendation model therefore combines context with explicit user preferences instead of relying solely on mood.

C. Emotion-Aware Music Recommendation

Emotion and contextual information have increasingly been incorporated into music recommendation research. Assuncao, Piccolo, and Zaina reviewed research involving emotional and contextual information in music recommendation and identified user preference, activity, satisfaction, cognitive load, and other contextual variables as relevant factors [5]. BusMind adopts this general principle while keeping its recommendation mechanism lightweight.

The detected expression is not directly treated as a psychological measurement. Instead, it is converted into an application-level state that can influence candidate music selection. This reduces the risk of overinterpreting the output of the facial-expression model.

D. Retrieval Practice and Spaced Learning

Retrieval practice involves recalling previously learned information rather than simply reviewing it. A systematic review of applied retrieval-practice research reported consistent benefits across different educational conditions [4]. Similarly, Carpenter, Pan, and Butler described spacing and retrieval practice as well-supported learning strategies across different contexts [3].

BusMind incorporates these principles into its flashcard system. Cards are stored with review history and can be assigned a priority according to recall performance, difficulty, and elapsed time since previous review.

E. Vision Transformers for Facial Expression Recognition

Vision Transformers apply the Transformer architecture to sequences of image patches. Dosovitskiy et al. demonstrated that image classification can be performed by representing an image as a sequence of fixed-size patches and processing the sequence through Transformer layers [6]. BusMind uses a Vision Transformer-based facial-expression model to obtain an expression category from an optional camera frame.

The use of a Transformer does not eliminate the limitations of facial-expression recognition. Model output remains dependent on image quality, facial visibility, training data, and environmental conditions.

F. Smartphone Inertial Sensing

Smartphones contain accelerometers and gyroscopes that provide measurements along the device axes. Reviews of smartphone-based human activity recognition show that inertial sensors are widely used for capturing movement and orientation-related information [7]. BusMind uses these sensors to provide posture-awareness feedback rather than attempting to perform clinical posture assessment.

G. Research Gap

The literature demonstrates that mobile learning, contextual recommendation, emotion-aware interaction, retrieval practice, and smartphone sensing have each been studied independently. The gap addressed by BusMind is the integration of these capabilities into a single commute-oriented learning workflow. The proposed system focuses on architectural integration, contextual activity selection, and practical mobile usability while avoiding unsupported claims about educational or medical outcomes.

III. PROPOSED SYSTEM

A. System Overview

BusMind is organized as a modular mobile application consisting of a presentation layer, state-management layer, service layer, persistence layer, and analytics layer. The application is implemented using Flutter and Dart, allowing the same application architecture to support mobile deployment while maintaining a consistent user interface.

The presentation layer provides the learner with access to authentication, learning activities, recommendations, flashcards, voice interaction, posture feedback, and analytics. Provider-based state management separates screen-level presentation from application logic.

Service components handle functions such as emotion inference, music recommendation, speech recognition, and sensor processing.

The persistence layer stores structured learning information locally. Hive is used for data requiring structured storage, while SharedPreferences is suitable for lightweight preferences and configuration values. This separation allows learning activities such as flashcard review to remain accessible even when network connectivity is temporarily unavailable.

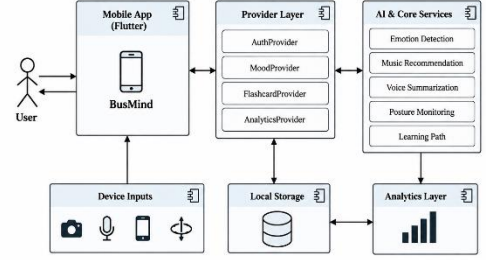


FIGURE 1. Overall architecture of the proposed BusMind mobile learning system.

B. Learner Context Model

The system represents learner context using profile, current-session, and historical information. Let the learner state be represented as

$$X = [X_p, X_c, X_h]$$

where (X_p) represents relatively stable profile information, (X_c) represents current contextual information, and (X_h) represents recent learning history.

The current context can be expressed as

$$X_c = [m, t, q, p, s]$$

where (m) denotes the detected application-level mood, (t) represents available session time, (q) represents topic priority, (p) represents user preference, and (s) represents the current session state.

This representation prevents a single signal from controlling the complete recommendation process.

IV. EMOTION-AWARE INTERACTION

A. Facial Expression Detection

The emotion module optionally captures a facial image using the smartphone camera. The image is passed to

the selected facial-expression recognition model, which produces probabilities for predefined expression categories. The model used in the prototype is based on a Vision Transformer and is configured for seven expression categories.

For an input image (I), the classifier produces

$$P(e_k | I), \quad k = 1, 2, \dots, K$$

where (K) represents the number of expression categories. The predicted expression is

$$e^* = \arg \max_{e_k} P(e_k | I)$$

]

The predicted expression is then converted into an application-level mood state.

Table I. Mapping of facial-expression categories to broad application-level contextual states.

Facial expression	Application context
Happy	Positive / energetic
Neutral	Neutral / focused
Sad	Low-energy / supportive
Angry	High-tension
Fear	Uneasy
Surprise	Alert
Disgust	Negative

The mapping is intentionally broad. It does not claim that a particular facial expression proves a user's psychological or emotional condition

This design follows the broader principle of contextual recommendation rather than making music selection depend entirely on facial-expression output.

For a candidate music group (g_j), a conceptual recommendation score is defined as

$$R(g_j) = w_m S_m + w_p S_p + w_t S_t + w_h S_h$$

where (S_m) represents mood compatibility, (S_p) represents user preference, (S_t) represents time compatibility, and (S_h) represents historical interaction. The weights (w_m, w_p, w_t, w_h) determine the relative contribution of each factor.

The selected candidate can then be represented as

$$g^* = \arg \max_{g_j} R(g_j)$$

This formulation also allows explicit user preferences to override an unsuitable recommendation generated from a noisy contextual signal.

V. ADAPTIVE LEARNING MODULE

A. Smart Flashcards

The flashcard component is designed around active recall. Each card contains learning content together with review metadata such as previous performance, difficulty, and last-review time.

The priority of a flashcard can be represented as

$$P_i = w_1(1 - A_i) + w_2 D_i + w_3 T_i + w_4 M_i$$

where (A_i) represents recent recall accuracy, (D_i) represents estimated difficulty, (T_i) represents normalized elapsed time, and (M_i) represents the learner's mastery gap.

A normalized time factor can be calculated using

$$T_i' = \frac{T_i - T_{\min}}{T_{\max} - T_{\min}}$$

A card with poor recall, high difficulty, or a longer review gap therefore receives greater priority.

Table II. Flashcard metadata used for adaptive review.

Parameter	Purpose
Difficulty	Represents perceived or assigned card difficulty
Recall history	Represents previous learner performance
Last review	Determines elapsed review time
Review count	Represents exposure frequency

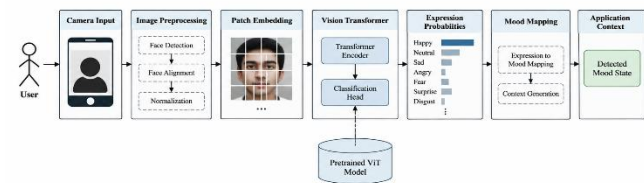


Fig. 2. Facial-expression detection and contextual mood generation pipeline.

FIGURE 2. Facial-expression processing and contextual mood generation pipeline.

B. Mood-Aware Music Recommendation

The music component combines the application-level mood with user preference and contextual information.

Mastery estimate	Represents current learning status
------------------	------------------------------------

The flashcard state is stored locally, allowing the learner to continue reviewing cards even when network access is unavailable.

B. Spaced Review

The review mechanism follows the principle that successfully recalled information can be revisited at increasing intervals, while incorrectly recalled information should receive earlier attention.

A simplified review interval can be expressed as

$$I_{next} = I_{current}(1 + \alpha C - \beta E)$$

where (C) represents successful recall, (E) represents an error signal, and (α) and (β) are tuning parameters.

This formulation is an application-level scheduling mechanism rather than an implementation of a complete established spaced-repetition algorithm.

C. Voice-Based Study Notes

Voice interaction is provided to reduce the amount of typing required during travel. The learner can speak a study-related note, after which speech recognition converts the audio into text.

The process can be represented as

$$A_{voice} \rightarrow T_{speech} \rightarrow T_{clean} \rightarrow S_{note}$$

where (A_{voice}) represents the audio signal, (T_{speech}) represents recognized speech, (T_{clean}) represents processed text, and (S_{note}) represents the generated study note.

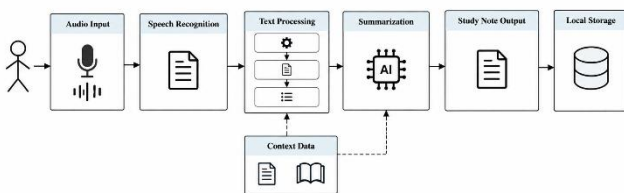


FIGURE 3. Voice-based study-note creation pipeline.

The separation of speech recognition and summarization allows either service to be improved independently.

VI. POSTURE-AWARE MOBILE SENSING

A. Sensor Model

The posture component uses accelerometer and gyroscope measurements. At time (t), the accelerometer vector is represented as

$$a(t) = [a_x, a_y, a_z]$$

and the gyroscope vector is represented as

$$g(t) = [g_x, g_y, g_z]$$

The combined sensor vector is

$$F(t) = [a_x, a_y, a_z, g_x, g_y, g_z]$$

The acceleration magnitude can be calculated as

$$A(t) = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

while the angular velocity magnitude can be calculated as

$$G(t) = \sqrt{g_x^2 + g_y^2 + g_z^2}$$

Rather than making decisions from individual readings, the system considers a temporal window

$$W_t = \{F(t - n + 1), \dots, F(t)\}$$

This reduces the influence of individual noisy measurements.

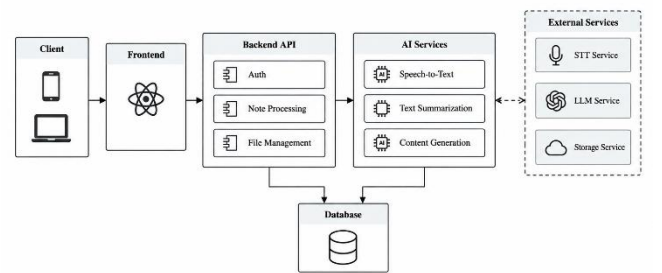


FIGURE 4. Sensor-processing architecture for posture-awareness feedback.

The current implementation is intended to provide awareness reminders and should not be interpreted as a medical or clinical posture assessment.

VII. CONTEXT-AWARE STUDY RECOMMENDATION

The central decision-making component of BusMind is the activity recommendation model. Instead of recommending an activity only from the learner's mood, the system combines multiple contextual factors.

Let the set of candidate activities be

$$A = \{a_1, a_2, \dots, a_n\}$$

For each activity, the system calculates

$$Score(a) = w_1L(a) + w_2C(a) + w_3P(a) + w_4M(a)$$

where $(L(a))$ represents learning value, $(C(a))$ represents contextual suitability, $(P(a))$ represents user preference, and $(M(a))$ represents mood compatibility.

The selected activity is

$$a^* = \arg \max_{a \in A} Score(a)$$

This design allows the application to recommend different activities according to the learner's available time and current learning state.

For example, if the remaining journey is short, a small flashcard session may receive a higher contextual score. If more time is available, the recommendation may shift toward voice-based note generation or a longer learning activity.

Table III. Contextual inputs used by the recommendation layer.

Context	Influence on recommendation
Available time	Determines feasible activity duration
Learning history	Identifies topics requiring review
User preference	Controls preferred activity types
Mood context	Provides an optional contextual signal
Topic priority	Gives importance to pending concepts
Current session	Determines ongoing activity state

VIII. GRAPH-BASED STUDY PATH

Learning topics are represented as a directed graph

$$G = (V, E)$$

where (V) represents topics and (E) represents prerequisite or recommended relationships.

A topic such as “Machine Learning” may depend on concepts such as probability, statistics, and linear algebra. Representing these dependencies as a graph allows BusMind to reason about the order in which topics can be studied.

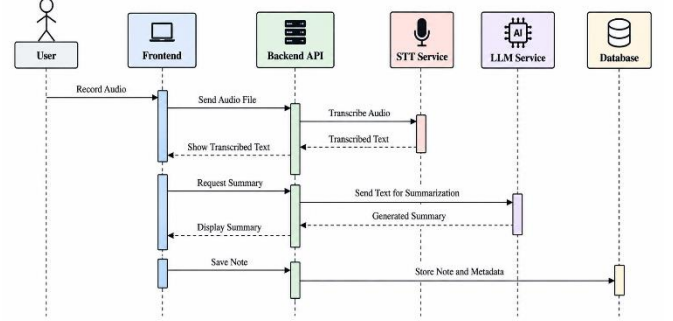


FIGURE 5. Graph-based representation of prerequisite relationships and study-path generation.

Breadth-first search can identify shallow prerequisite relationships, while depth-first search can explore deeper dependency chains. A* can incorporate estimated study cost when a preferred path must be selected.

The A* evaluation function is

$$f(n) = g(n) + h(n)$$

where $(g(n))$ represents accumulated cost and $(h(n))$ represents estimated remaining cost.

A learning-path cost can be defined as

$$g(n) = w_1T_n + w_2D_n + w_3R_n$$

where (T_n) is estimated study time, (D_n) is topic difficulty, and (R_n) is review priority.

This allows the path generator to prefer topics that are both educationally relevant and practical within the learner's available time.

IX. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Provider-Based Architecture

BusMind uses Provider-based state management to separate interface components from application logic. Authentication, mood detection, music, flashcards, voice interaction, posture monitoring, and analytics are represented as separate logical providers.

Table IV. Major application providers and their responsibilities.

Provider	Responsibility
AuthProvider	Authentication and session state
AppStateProvider	Global application state
MoodDetectionProvider	Expression and mood state
MusicProvider	Recommendation and playback
FlashcardProvider	Flashcard state and review
VoiceProvider	Speech and note generation
PostureProvider	Sensor processing
AnalyticsProvider	Learning-event aggregation

This architecture allows individual services to be modified without requiring substantial changes to unrelated parts of the application.

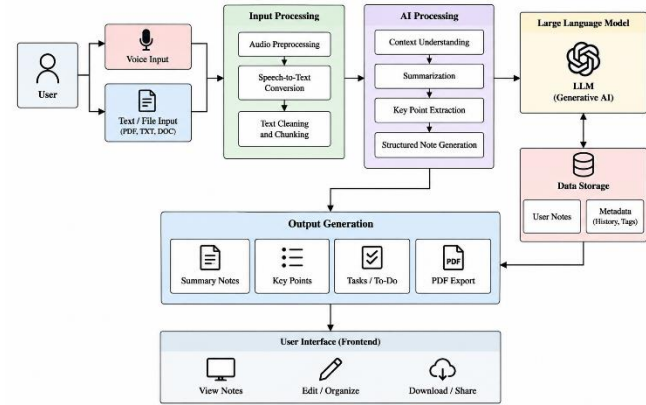


FIGURE 6. Provider-based software architecture of *BusMind*.

B. Local Persistence and Offline Operation

Commuting environments can involve intermittent network connectivity. *BusMind* therefore separates network-dependent services from locally available learning functions. Flashcard information, review history, preferences, and selected analytics can remain accessible through local persistence.

Hive is used for structured application information, while lightweight configuration values can be maintained through `SharedPreferences`. The

architecture follows a graceful-degradation principle in which the failure of an optional service does not disable the entire application.

For example, if camera access is denied, the learner can continue using flashcards and voice interaction. If an external AI service is temporarily unavailable, locally stored learning content can still be accessed.

C. Data Flow

The overall data flow can be represented as

$$\begin{aligned} \text{Input} &\rightarrow \text{Processing} \rightarrow \text{Context} \\ &\rightarrow \text{Recommendation} \rightarrow \text{Activity} \\ &\rightarrow \text{Feedback} \end{aligned}$$

The feedback generated from the activity is stored as part of the learner's history. This history then becomes an input for subsequent recommendations.

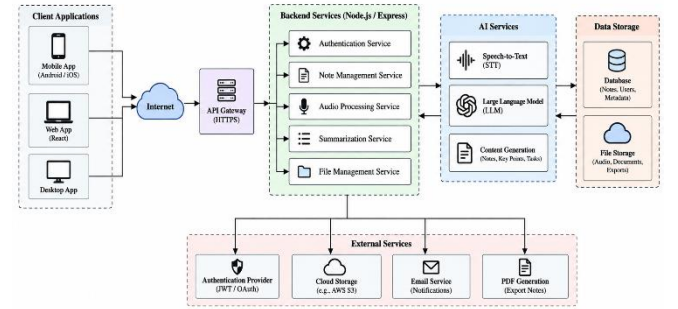


FIGURE 7. End-to-end data flow and feedback loop in *BusMind*.

X. SECURITY, PRIVACY, AND RESPONSIBLE AI

BusMind processes several types of potentially sensitive information, including facial images, voice input, learning history, and sensor information. The system should therefore follow data-minimization principles and collect only information required for the selected functionality.

Camera and microphone features should remain permission-controlled. Raw images and audio should not be retained unnecessarily when derived information is sufficient for the application. Network communication should use secure transport, while authentication information should be protected using appropriate platform mechanisms.

Emotion recognition requires particular caution. A facial expression should not be presented as proof of a psychological state, mental-health condition, or personality characteristic. *BusMind* therefore uses expression recognition only as a contextual input to recommendation.

Similarly, posture feedback is intended to increase awareness of device orientation and prolonged patterns. It should not be represented as medical diagnosis, rehabilitation guidance, or clinical assessment.

XI. EXPERIMENTAL EVALUATION

The current prototype is evaluated primarily at the functional level because the main objective is to demonstrate integration of the proposed modules. Each component can be verified by examining whether its expected input is correctly transformed into the required output.

Table V. Functional verification of the major BusMind modules.

Module	Input	Expected output
Authentication	User credentials	Valid application session
Emotion detection	Camera image	Expression and contextual mood
Music recommendation	Mood and preferences	Candidate recommendation
Flashcards	Question and review response	Updated review state
Voice interaction	Spoken input	Recognized text
Posture monitoring	Sensor sequence	Awareness feedback
Analytics	Activity event	Updated progress information
Study path	Topic graph	Recommended topic sequence

The evaluation can be extended into quantitative testing once sufficient user interaction data is collected. For flashcards, recall accuracy can be represented as

$$Accuracy_{recall} = \frac{Correct\ Responses}{Total\ Responses} \times 100$$

The activity completion rate can be expressed as

$$Completion = \frac{Completed\ Activities}{Started\ Activities} \times 100$$

For delayed evaluation, retention can be measured as

$$Retention = \frac{Delayed\ Correct\ Responses}{Delayed\ Questions} \times 100$$

For emotion classification, standard multi-class metrics such as precision, recall, and F1-score can be used.

$$Precision = \frac{TP}{TP + FP}$$

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

These measures should be calculated using an appropriate labeled dataset and should not be replaced with fabricated prototype accuracy values.

XII. RESULTS AND DISCUSSION

The implemented prototype demonstrates the feasibility of combining learning, recommendation, sensing, and AI services within a single mobile application. The modular design allows individual components to operate without tightly coupling the complete system to one service.

The emotion module provides a contextual signal that can influence recommendations, while the music module combines that signal with user preferences. The flashcard module supports retrieval-based learning and records review information locally. Voice interaction reduces the need for typing during travel, while sensor processing provides posture-awareness feedback.

A major advantage of the architecture is its ability to handle different levels of connectivity. Core learning functions do not need to depend continuously on external services. This is particularly important for a bus-oriented application because network quality can vary during travel.

The recommendation architecture also avoids making a single prediction responsible for the learner's complete experience. Mood, learning history, time availability, and preference are considered together. This makes the system more robust conceptually and provides a clearer path toward future personalization.

However, the current implementation should not be interpreted as evidence that BusMind improves academic performance or wellbeing. Such claims require controlled experiments involving sufficient participants, standardized learning material, appropriate comparison groups, and statistically meaningful analysis.

XIII. LIMITATIONS

The primary limitation of the current system is that several components rely on heuristic or contextual decisions rather than extensively trained personalized models. Facial-expression recognition can be affected by lighting, camera position, facial orientation, and differences between training data and real-world users. Consequently, the detected expression should be treated as an uncertain contextual signal.

The recommendation weights also require empirical calibration. The current formulation provides a framework for combining contextual information but does not claim that the selected weights are universally optimal.

Voice interaction can introduce speech-recognition errors, particularly in noisy bus environments. Summarization quality can also vary according to the clarity and technical complexity of the spoken content.

Posture monitoring is similarly affected by smartphone position. A phone held in different orientations can produce substantially different sensor measurements even when the learner's physical posture remains similar. The current module therefore provides awareness-oriented feedback rather than medical assessment.

Finally, the current prototype does not include a controlled longitudinal study. The educational effectiveness, sustained engagement, and personalization quality of BusMind therefore remain subjects for future empirical evaluation.

XIV. FUTURE WORK

Future development will focus on strengthening the personalization layer through real interaction data. Instead of using manually selected recommendation weights, future versions can learn preference patterns from user feedback while maintaining explicit controls that allow learners to override recommendations.

The emotion component can be improved by evaluating different facial-expression models under realistic mobile conditions. Privacy-preserving approaches could also be investigated so that raw facial images do not need to leave the device.

The learning engine can be extended by incorporating stronger spaced-repetition models and learner mastery estimation. The graph-based learning component can also be connected with mastery information so that prerequisite paths are generated according to actual learner weaknesses.

Posture monitoring can be improved by collecting annotated sensor sequences and training a lightweight classification model. This could reduce dependence on manually selected thresholds while preserving the application's low-resource design.

A future controlled study should compare BusMind with a conventional mobile-learning application. The study can measure recall, delayed retention, activity completion, session consistency, perceived usefulness, and user satisfaction. Such an evaluation would provide evidence for determining whether contextual personalization produces measurable benefits.

The system can also be extended to other fragmented learning environments, such as metro journeys, waiting periods, campus travel, and short breaks between academic activities.

XV. CONCLUSION

BusMind presents a context-aware mobile learning architecture designed specifically for students who spend significant amounts of time travelling by bus. The proposed system combines facial-expression-based contextual sensing, mood-aware music recommendation, adaptive flashcards, voice-based study notes, posture awareness, analytics, and graph-based study-path generation within a unified Flutter application.

The system is designed around the observation that commuting does not provide the same learning conditions as a conventional study environment. Available time, attention, connectivity, and user state can change continuously. BusMind therefore treats these factors as contextual information that can influence activity selection rather than assuming a fixed learning sequence.

The integration of retrieval practice and spaced review provides an established learning foundation for the flashcard component, while contextual recommendation allows the application to adapt its interactions according to the learner's current situation. Smartphone sensors and optional facial-expression analysis further expand the range of contextual information available to the system.

The architecture also emphasizes modularity, local persistence, graceful degradation, and responsible use of AI. In particular, the system avoids treating facial expressions as psychological diagnoses or posture feedback as medical assessment.

The current work should be considered a functional prototype and architectural contribution. Its effectiveness as a long-term educational system remains

to be established through controlled experimentation and real-world user studies. Nevertheless, the proposed architecture provides a practical foundation for combining AI-assisted personalization, mobile learning, and contextual sensing within fragmented daily travel time.

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